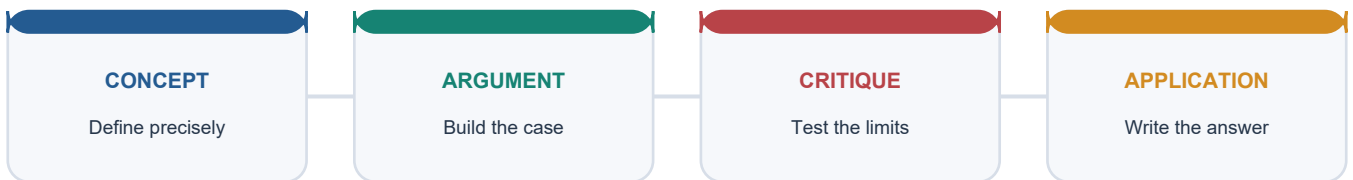


COMPLETE LEARNING SESSION

Irrigation, Inputs, Credit, Insurance and Sustainable Agriculture - Complete Learning Session

Economy | UPSC Prelims and GS-III

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / SESSION INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Irrigation, Inputs, Credit, Insurance and Sustainable Agriculture - Complete Learning Session . . . 5

SOURCE, STATUS AND DATE CONTROL 5

OFFICIAL LINKS CHECKED 5

ORIGIN AND TIMELINE 5

CURRENT OFFICIAL DASHBOARD 6

BASIC LEARNING SESSION 6

CORE SESSION 1 - Water-input-credit-insurance nexus 6

CORE SESSION 2 - Irrigation potential and utilisation 7

CORE SESSION 3 - Net, gross irrigated area and intensity 8

CORE SESSION 4 - Canal irrigation 8

CORE SESSION 5 - Groundwater irrigation 9

CORE SESSION 6 - Tank and lift irrigation 10

CORE SESSION 7 - Command area and CADWM 11

CORE SESSION 8 - Watershed development 12

CORE SESSION 9 - Micro-irrigation 13

CORE SESSION 10 - Irrigation-efficiency layers 14

CORE SESSION 11 - Tensiometer and irrigation scheduling 15

CORE SESSION 12 - Rebound effect in water use 16

CORE SESSION 13 - Groundwater-energy-crop nexus 17

CORE SESSION 14 - Canal externalities and drainage 18

CORE SESSION 15 - Water pricing 19

CORE SESSION 16 - Participatory irrigation management 20

CORE SESSION 17 - PMKSY current architecture 21

CORE SESSION 18 - AIBP and Har Khet Ko Pani 22

CORE SESSION 19 - Per Drop More Crop current location 23

CORE SESSION 20 - Seed multiplication chain 24

CORE SESSION 21 - Formal and informal seed systems 25

CORE SESSION 22 - Seed law and current Bill status 26

CORE SESSION 23 - Seed and planting-material diversity 27

CORE SESSION 24 - N, P and K functions	28
CORE SESSION 25 - Nutrient imbalance	29
CORE SESSION 26 - Urea control and NBS	30
CORE SESSION 27 - Fertiliser DBT	31
CORE SESSION 28 - Soil Health Card and INM	32
CORE SESSION 29 - Biochar and soil amendments	33
CORE SESSION 30 - Nano-fertiliser evidence discipline	34
CORE SESSION 31 - Pesticides and IPM	35
CORE SESSION 32 - Mechanisation	36
CORE SESSION 33 - Custom hiring centres	37
CORE SESSION 34 - Credit need and sources	38
CORE SESSION 35 - Commercial banks, RRBs and cooperatives	39
CORE SESSION 36 - NABARD role	40
CORE SESSION 37 - Priority-sector agriculture target	41
CORE SESSION 38 - Kisan Credit Card	42
CORE SESSION 39 - MISS and current KCC limits	43
CORE SESSION 40 - Tenant, JLG and SHG inclusion	44
CORE SESSION 41 - Insurance principles	45
CORE SESSION 42 - Indemnity, yield and weather index	46
CORE SESSION 43 - Area approach and basis risk	47
CORE SESSION 44 - PMFBY current architecture	48
CORE SESSION 45 - PMFBY loss stages and technology	49
CORE SESSION 46 - Risk layering	50
CORE SESSION 47 - Conservation agriculture	51
CORE SESSION 48 - Integrated nutrient and pest management	52
CORE SESSION 49 - Organic, natural and permaculture	53
CORE SESSION 50 - Natural Farming Mission current status	54
CORE SESSION 51 - Sikkim organic transition	54
CORE SESSION 52 - Agroforestry, millets and integrated farming	55
CORE SESSION 53 - Climate-smart and climate-resilient agriculture	56
CORE SESSION 54 - Precision and digital agriculture boundary	57
CORE SESSION 55 - Input-subsidy externalities	58

CORE SESSION 56 - Integrated productivity-equity-sustainability reform	59
BASIC MCQS / REMEDIATION	60
PYQS AND ANSWER PRACTICE	60
OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER	61
ADVANCED 1 - Common-pool groundwater	61
ADVANCED 2 - Jevons-style irrigation rebound	61
ADVANCED 3 - Credit rationing	61
ADVANCED 4 - Basis-risk optimisation	61
ADVANCED 5 - Subsidy incidence	61
ADVANCED 6 - Adaptation portfolios	61
CONSOLIDATED REGISTER NOTES	62
1. Irrigation accounting and types	62
2. PMKSY and governance	62
3. Seeds	62
4. Fertilisers, soil and pests	62
5. Machinery and credit	62
6. Insurance	63
7. Sustainable agriculture	63
8. Reform architecture	63

Irrigation, Inputs, Credit, Insurance and Sustainable Agriculture - Complete Learning Session

SOURCE, STATUS AND DATE CONTROL

- **Current-source cutoff:** 9 September 2026.
- **Source order:** canonical Economy/Agriculture/Environment Basic/Core and Optional Advanced Markdown; OCR-derived Ramesh Singh and *Economic Survey 2025-26* notes; official Agriculture, Jal Shakti, Land Resources and Fertilizers ministries, RBI, NABARD, PMKSY, PMFBY, Soil Health, ICAR and PIB material.
- **Scheme rule:** launch, approval period, component location, budget announcement, operating guideline, enrolment, claim and outcome remain separate.
- **Threshold rule:** every premium, credit limit, priority-sector target and subsidy architecture carries its date, denominator and qualification.
- **Boundary:** cropping-pattern detail belongs to Topic 11; market architecture to Topic 13; digital-agriculture governance to Topic 27.

OFFICIAL LINKS CHECKED

- PMKSY official objectives: <https://www.pmkSY.gov.in/AboutPMKSY.aspx>
- PIB PMKSY 2026 backgrounder: <https://static.pib.gov.in/WriteReadData/specificdocs/documents/2026/jun/doc2026630906701.pdf>
- Agriculture Ministry Seeds Act/Rules and draft Seeds Bill, 2025: <https://agriwelfare.gov.in/en/SeedsDiv>
- Department of Fertilizers subsidy architecture: <https://fert.gov.in/en/department/our-wings/fertilizer-subsidy>
- Soil Health Card resources: <https://soilhealth.dac.gov.in/GeneralResources>
- RBI PSL Directions, 2025, updated 7 August 2026: <https://www.rbi.org.in/Scripts/NotificationUser.aspx?Id=12799>
- PIB/RBI MISS FY2025-26: <https://pib.gov.in/PressReleasePage.aspx?PRID=2131989>
- PMFBY official portal and PIB August 2026 backgrounder: <https://pmfby.gov.in/>
- ICAR crop, soil and IPM research: <https://icar.gov.in/>

ORIGIN AND TIMELINE

Date	Development	Current-status qualification
1966 / 1968	Seeds Act / Seeds Rules	Core seed-quality law remains in force at cutoff
1968 / 1971	Insecticides Act / Rules	Core pesticide-control framework
1982	NABARD established	Apex rural development finance institution
1998	Kisan Credit Card introduced	Credit product; interest support follows separate scheme
2015	PMKSY and Soil Health Card launched	Component locations have since changed
2016	PMFBY launched	Farmer enrolment voluntary since Kharif 2020
2021-26	Approved PMKSY continuation window	Official 2026 material continues umbrella context
2022-23	PDMC and Soil Health/Fertility placed under RKVY	Current administrative location
25 November 2024	National Mission on Natural Farming approved	Approved through 2025-26; later status needs fresh authority

Date	Development	Current-status qualification
March 2025 / 7 August 2026	RBI PSL Directions issued / updated	Agriculture 18%; NCF 14%; SMF 10% for applicable banks
November-December 2025	Draft Seeds Bill, 2025 consultation	Draft, not enacted by cutoff
August 2026	PMFBY official update	2% / 1.5% / 5% farmer premium structure retained

CURRENT OFFICIAL DASHBOARD

Item	Dated status	Qualification
PMKSY	2021-26 approved umbrella; official June 2026 backgrounder continues programme context	PDMC is under RKVY from 2022-23
Seeds	Seeds Act 1966/Rules 1968 remain; 2025 Bill was a consultation draft	Draft != enacted Act
Fertiliser	Urea controlled; eligible P&K under NBS; DBT pays companies after PoS sale	Seasonal NBS rates are not frozen here
PSL	RBI 2025 Directions updated 7 August 2026	18% agriculture; nested 14% NCF and 10% SMF for applicable banks
KCC/MISS	Budget 2025-26 announced Rs 5 lakh overall KCC limit; FY2025-26 MISS retained concessional slab up to Rs 3 lakh	4% effective rate requires prompt repayment
PMFBY	August 2026 official terms retain farmer shares 2% Kharif / 1.5% Rabi food-oilseed / 5% annual commercial-horticultural	Voluntary for farmers since Kharif 2020; specified risks only

BASIC LEARNING SESSION

CORE SESSION 1 - Water-input-credit-insurance nexus

VISUAL FIRST

SOIL + WATER + SEED + NUTRIENTS + KNOWLEDGE
 -> CREDIT-FINANCED CULTIVATION -> RISK -> INSURANCE / ADAPTATION

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Farm productivity is produced by a linked system of natural resources, inputs, finance, knowledge and risk management.

ANSWER-GRABBING LINE: Isolated subsidies can relax one farm constraint while worsening another resource or fiscal constraint.

MUST-WRITE KEYWORDS: water-input-credit-insurance, nexus, isolated, subsidies, relax, farm, constraint, while, worsening, another, resource.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Isolated subsidies can relax one farm constraint while worsening another resource or fiscal constraint.
- **Named evidence:** The canonical Economy and Agriculture owners organise this topic through the water-energy-input-credit-insurance nexus.
- **Analysis:** Reliable water raises the return to seed and fertiliser, while timely credit determines whether those inputs can be purchased.
- **Qualification / demand link:** Market access and cropping pattern remain separate but interacting topics.

EVIDENCE AND INFERENCE

- **Fact/evidence:** The canonical Economy and Agriculture owners organise this topic through the water-energy-input-credit-insurance nexus.
- **Analytical inference:** Reliable water raises the return to seed and fertiliser, while timely credit determines whether those inputs can be purchased.
- **Qualification:** Market access and cropping pattern remain separate but interacting topics.

PRELIMS TRAP

Do not attribute productivity to one input without the complementary-input chain.

MAINS USE

Use the nexus to frame productivity, equity and sustainability together.

MINI RECAP

- Definition: Farm productivity is produced by a linked system of natural resources, inputs, finance, knowledge and risk management.
- Evidence anchor: The canonical Economy and Agriculture owners organise this topic through the water-energy-input-credit-insurance nexus.
- Limitation: Market access and cropping pattern remain separate but interacting topics.

CORE SESSION 2 - Irrigation potential and utilisation

VISUAL FIRST

ULTIMATE POTENTIAL -> POTENTIAL CREATED -> POTENTIAL UTILISED
-> RELIABLE FIELD DELIVERY -> CROP OUTPUT

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Irrigation potential measures designed capacity, while utilisation measures the area actually served under the relevant system.

ANSWER-GRABBING LINE: Created capacity is not evidence of reliable, equitable or productive field delivery.

MUST-WRITE KEYWORDS: irrigation, potential, utilisation, created, capacity, evidence, reliable, equitable, productive, field, delivery.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Created capacity is not evidence of reliable, equitable or productive field delivery.
- **Named evidence:** PMKSY and command-area programmes distinguish project creation from completion and utilisation.
- **Analysis:** Maintenance, field channels, scheduling and tail-end access determine whether created potential becomes farm water.
- **Qualification / demand link:** Area served still does not measure water-use efficiency.

EVIDENCE AND INFERENCE

- **Fact/evidence:** PMKSY and command-area programmes distinguish project creation from completion and utilisation.
- **Analytical inference:** Maintenance, field channels, scheduling and tail-end access determine whether created potential becomes farm water.
- **Qualification:** Area served still does not measure water-use efficiency.

PRELIMS TRAP

Do not merge potential created, potential utilised and irrigated area.

MAINS USE

Separate infrastructure stock from service flow and outcome.

MINI RECAP

- Definition: Irrigation potential measures designed capacity, while utilisation measures the area actually served under the relevant system.
- Evidence anchor: PMKSY and command-area programmes distinguish project creation from completion and utilisation.
- Limitation: Area served still does not measure water-use efficiency.

CORE SESSION 3 - Net, gross irrigated area and intensity

VISUAL FIRST

NET IRRIGATED AREA = LAND IRRIGATED AT LEAST ONCE
GROSS IRRIGATED AREA = ALL IRRIGATED CROP INSTANCES
IRRIGATION INTENSITY = $\text{GROSS} / \text{NET} \times 100$

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Net irrigated area counts land; gross irrigated area counts repeated irrigated crop use over the year.

ANSWER-GRABBING LINE: Irrigation intensity captures multiple irrigated cropping, not efficiency per unit of water.

MUST-WRITE KEYWORDS: gross, irrigated, area, intensity, irrigation, captures, multiple, cropping, efficiency, unit, water.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Irrigation intensity captures multiple irrigated cropping, not efficiency per unit of water.
- **Named evidence:** Official agricultural statistics distinguish net and gross areas before calculating intensity.
- **Analysis:** Multiple seasons can raise gross area without expanding the physical land base.
- **Qualification / demand link:** A high ratio can coexist with groundwater stress or low application efficiency.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Official agricultural statistics distinguish net and gross areas before calculating intensity.
- **Analytical inference:** Multiple seasons can raise gross area without expanding the physical land base.
- **Qualification:** A high ratio can coexist with groundwater stress or low application efficiency.

PRELIMS TRAP

Do not use irrigation intensity as a water-productivity measure.

MAINS USE

State numerator, denominator and unit in Prelims and Mains.

MINI RECAP

- Definition: Net irrigated area counts land; gross irrigated area counts repeated irrigated crop use over the year.
- Evidence anchor: Official agricultural statistics distinguish net and gross areas before calculating intensity.
- Limitation: A high ratio can coexist with groundwater stress or low application efficiency.

CORE SESSION 4 - Canal irrigation

VISUAL FIRST

RESERVOIR / BARRAGE -> MAIN CANAL -> DISTRIBUTARY
-> MINOR / FIELD CHANNEL -> HEAD / MIDDLE / TAIL FARM

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Canal irrigation conveys surface water through a network to a command area.

ANSWER-GRABBING LINE: Canals spread fixed infrastructure benefits but create distribution, maintenance and ecological externalities.

MUST-WRITE KEYWORDS: canal, irrigation, canals, spread, fixed, infrastructure, benefits, create, distribution, maintenance, ecological.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Canals spread fixed infrastructure benefits but create distribution, maintenance and ecological externalities.
- **Named evidence:** AIBP and command-area programmes are the principal Union support anchors for major and medium systems.
- **Analysis:** Gravity delivery can serve many farms and reduce individual pumping dependence.
- **Qualification / demand link:** Seepage, siltation, waterlogging, salinity and head-tail inequity vary by system.

EVIDENCE AND INFERENCE

- **Fact/evidence:** AIBP and command-area programmes are the principal Union support anchors for major and medium systems.
- **Analytical inference:** Gravity delivery can serve many farms and reduce individual pumping dependence.
- **Qualification:** Seepage, siltation, waterlogging, salinity and head-tail inequity vary by system.

PRELIMS TRAP

Do not treat seepage as only loss; it may also recharge aquifers.

MAINS USE

Use conveyance, distribution and drainage as separate reform layers.

MINI RECAP

- Definition: Canal irrigation conveys surface water through a network to a command area.
- Evidence anchor: AIBP and command-area programmes are the principal Union support anchors for major and medium systems.
- Limitation: Seepage, siltation, waterlogging, salinity and head-tail inequity vary by system.

CORE SESSION 5 - Groundwater irrigation

VISUAL FIRST

RECHARGE -> AQUIFER STOCK -> WELL / TUBEWELL PUMPING
-> PRIVATE FLEXIBILITY -> CUMULATIVE DEPLETION RISK

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Groundwater irrigation draws stored subsurface water through wells and tubewells.

ANSWER-GRABBING LINE: Private control over pumping creates flexibility, but the aquifer is a shared common-pool resource.

MUST-WRITE KEYWORDS: groundwater, irrigation, private, control, over, pumping, creates, flexibility, aquifer, shared, common-pool.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Private control over pumping creates flexibility, but the aquifer is a shared common-pool resource.
- **Named evidence:** Central Ground Water Board assessments and Atal Bhujal Yojana anchor aquifer-status and community-management analysis.
- **Analysis:** On-demand pumping supports crop timing and reduces canal-schedule dependence.
- **Qualification / demand link:** Cheap power, water-intensive crops and weak measurement can exceed recharge.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Central Ground Water Board assessments and Atal Bhujal Yojana anchor aquifer-status and community-management analysis.
- **Analytical inference:** On-demand pumping supports crop timing and reduces canal-schedule dependence.
- **Qualification:** Cheap power, water-intensive crops and weak measurement can exceed recharge.

PRELIMS TRAP

Do not infer sustainable supply from the existence of a functioning borewell.

MAINS USE

Link individual incentives to aquifer-level water budgets.

MINI RECAP

- Definition: Groundwater irrigation draws stored subsurface water through wells and tubewells.
- Evidence anchor: Central Ground Water Board assessments and Atal Bhujal Yojana anchor aquifer-status and community-management analysis.
- Limitation: Cheap power, water-intensive crops and weak measurement can exceed recharge.

CORE SESSION 6 - Tank and lift irrigation

VISUAL FIRST

LOCAL STORAGE / RIVER -> TANK OR PUMPED LIFT
-> DISTRIBUTION -> FARM
ENERGY + MAINTENANCE CONDITION RELIABILITY

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Tank irrigation stores local runoff, while lift irrigation uses energy to raise water from a lower source.

ANSWER-GRABBING LINE: Decentralised sources can suit local terrain but require catchment, energy and maintenance governance.

MUST-WRITE KEYWORDS: tank, lift, irrigation, decentralised, sources, suit, local, terrain, require, catchment, energy.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Decentralised sources can suit local terrain but require catchment, energy and maintenance governance.
- **Named evidence:** HKKP includes surface minor irrigation and repair, renovation and restoration of water bodies.
- **Analysis:** Local systems can reduce long conveyance and support supplementary irrigation.
- **Qualification / demand link:** Encroachment, siltation, pump cost and source depletion can reduce service.

EVIDENCE AND INFERENCE

- **Fact/evidence:** HKKP includes surface minor irrigation and repair, renovation and restoration of water bodies.
- **Analytical inference:** Local systems can reduce long conveyance and support supplementary irrigation.
- **Qualification:** Encroachment, siltation, pump cost and source depletion can reduce service.

PRELIMS TRAP

Do not classify every pumped well as a public lift-irrigation project.

MAINS USE

Match source, terrain, energy and institution.

MINI RECAP

- Definition: Tank irrigation stores local runoff, while lift irrigation uses energy to raise water from a lower source.
- Evidence anchor: HKKP includes surface minor irrigation and repair, renovation and restoration of water bodies.
- Limitation: Encroachment, siltation, pump cost and source depletion can reduce service.

CORE SESSION 7 - Command area and CADWM

VISUAL FIRST

PROJECT OUTLET -> COMMAND AREA -> FIELD CHANNEL
-> ON-FARM DEVELOPMENT -> ROTATIONAL / MEASURED DELIVERY

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

A command area is the land designed to receive irrigation from a project; CADWM improves last-mile distribution and field use.

ANSWER-GRABBING LINE: Project completion without command-area works can leave a gap between potential and utilisation.

MUST-WRITE KEYWORDS: command, area, cadwm, project, completion, without, command-area, works, leave, between, potential.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Project completion without command-area works can leave a gap between potential and utilisation.
- **Named evidence:** Command Area Development and Water Management is linked with AIBP/HKKP architecture.
- **Analysis:** Lined channels, field drains, measurement and scheduling can improve equity and utilisation.
- **Qualification / demand link:** Engineering upgrades fail without maintenance and farmer coordination.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Command Area Development and Water Management is linked with AIBP/HKKP architecture.
- **Analytical inference:** Lined channels, field drains, measurement and scheduling can improve equity and utilisation.
- **Qualification:** Engineering upgrades fail without maintenance and farmer coordination.

PRELIMS TRAP

Do not confuse command area with watershed catchment.

MAINS USE

Use CADWM to connect dam or canal assets with field outcomes.

MINI RECAP

- Definition: A command area is the land designed to receive irrigation from a project; CADWM improves last-mile distribution and field use.
- Evidence anchor: Command Area Development and Water Management is linked with AIBP/HKKP architecture.
- Limitation: Engineering upgrades fail without maintenance and farmer coordination.

CORE SESSION 8 - Watershed development

VISUAL FIRST

RIDGE -> CONTOUR / VEGETATION -> CHECK STRUCTURES
-> RECHARGE / MOISTURE -> VALLEY LIVELIHOODS

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

A watershed is a geo-hydrological area draining to a common outlet and treated from ridge to valley.

ANSWER-GRABBING LINE: Watershed development manages runoff, soil moisture, recharge and livelihoods across a catchment rather than one plot.

MUST-WRITE KEYWORDS: watershed, development, manages, runoff, soil, moisture, recharge, livelihoods, across, catchment, rather.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Watershed development manages runoff, soil moisture, recharge and livelihoods across a catchment rather than one plot.
- **Named evidence:** WDC-PMKSY under the Department of Land Resources carries forward the earlier IWMP approach.
- **Analysis:** Sequenced treatment reduces erosion and improves protective water in rainfed areas.
- **Qualification / demand link:** Benefits depend on tenure, participation, maintenance and downstream effects.

EVIDENCE AND INFERENCE

- **Fact/evidence:** WDC-PMKSY under the Department of Land Resources carries forward the earlier IWMP approach.
- **Analytical inference:** Sequenced treatment reduces erosion and improves protective water in rainfed areas.
- **Qualification:** Benefits depend on tenure, participation, maintenance and downstream effects.

PRELIMS TRAP

Do not reduce watershed development to building one check dam.

MAINS USE

Use catchment unit, ridge-to-valley sequence and user institutions.

MINI RECAP

- Definition: A watershed is a geo-hydrological area draining to a common outlet and treated from ridge to valley.
- Evidence anchor: WDC-PMKSY under the Department of Land Resources carries forward the earlier IWMP approach.
- Limitation: Benefits depend on tenure, participation, maintenance and downstream effects.

CORE SESSION 9 - Micro-irrigation

VISUAL FIRST

SOURCE -> FILTER / PRESSURE -> DRIP LINE OR SPRINKLER
-> PRECISE ROOT-ZONE / CROP APPLICATION

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Micro-irrigation applies water through drip or sprinkler systems with greater spatial and temporal control than uncontrolled flooding.

ANSWER-GRABBING LINE: Precision can improve field application and fertigation, but adoption and maintenance determine realised gains.

MUST-WRITE KEYWORDS: micro-irrigation, precision, improve, field, application, fertigation, adoption, maintenance, determine, realised, gains.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Precision can improve field application and fertigation, but adoption and maintenance determine realised gains.
- **Named evidence:** Per Drop More Crop supports micro-irrigation; from 2022-23 it is implemented under RKVY rather than the core PMKSY umbrella.
- **Analysis:** Lower conveyance and evaporation losses can improve output per unit applied.
- **Qualification / demand link:** Capital cost, clogging, energy, water quality and crop suitability limit adoption.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Per Drop More Crop supports micro-irrigation; from 2022-23 it is implemented under RKVY rather than the core PMKSY umbrella.
- **Analytical inference:** Lower conveyance and evaporation losses can improve output per unit applied.
- **Qualification:** Capital cost, clogging, energy, water quality and crop suitability limit adoption.

PRELIMS TRAP

Do not say every sprinkler delivers water only at the root zone.

MAINS USE

Distinguish drip, sprinkler and their suitable contexts.

MINI RECAP

- Definition: Micro-irrigation applies water through drip or sprinkler systems with greater spatial and temporal control than uncontrolled flooding.
- Evidence anchor: Per Drop More Crop supports micro-irrigation; from 2022-23 it is implemented under RKVY rather than the core PMKSY umbrella.
- Limitation: Capital cost, clogging, energy, water quality and crop suitability limit adoption.

CORE SESSION 10 - Irrigation-efficiency layers

VISUAL FIRST

CONVEYANCE EFFICIENCY -> DISTRIBUTION EFFICIENCY
-> APPLICATION EFFICIENCY -> CROP WATER PRODUCTIVITY

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Irrigation efficiency can refer to useful delivery relative to withdrawal, useful application relative to field supply, or output per unit water.

ANSWER-GRABBING LINE: The denominator must be specified before comparing technologies or regions.

MUST-WRITE KEYWORDS: irrigation-efficiency, layers, denominator, must, specified, before, comparing, technologies, regions.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** The denominator must be specified before comparing technologies or regions.
- **Named evidence:** PMKSY's efficiency objective and micro-irrigation programmes focus on on-farm use, while basin accounting asks a wider question.
- **Analysis:** Reducing non-beneficial evaporation can generate real saving.
- **Qualification / demand link:** Return flows counted as field loss may recharge or serve downstream users.

EVIDENCE AND INFERENCE

- **Fact/evidence:** PMKSY's efficiency objective and micro-irrigation programmes focus on on-farm use, while basin accounting asks a wider question.
- **Analytical inference:** Reducing non-beneficial evaporation can generate real saving.
- **Qualification:** Return flows counted as field loss may recharge or serve downstream users.

PRELIMS TRAP

Do not use 'efficiency' without specifying scale and denominator.

MAINS USE

Write conveyance, application, consumptive use and productivity separately.

MINI RECAP

- Definition: Irrigation efficiency can refer to useful delivery relative to withdrawal, useful application relative to field supply, or output per unit water.
- Evidence anchor: PMKSY's efficiency objective and micro-irrigation programmes focus on on-farm use, while basin accounting asks a wider question.
- Limitation: Return flows counted as field loss may recharge or serve downstream users.

CORE SESSION 11 - Tensiometer and irrigation scheduling

VISUAL FIRST

SOIL-MOISTURE TENSION -> THRESHOLD -> IRRIGATE
-> AVOID TOO EARLY / TOO LATE APPLICATION

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

A tensiometer measures soil-water tension and helps schedule irrigation in its suitable measurement range.

ANSWER-GRABBING LINE: Scheduling can save water only when the farmer can control delivery timing and quantity.

MUST-WRITE KEYWORDS: tensiometer, irrigation, scheduling, save, water, only, when, farmer, control, delivery, timing.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Scheduling can save water only when the farmer can control delivery timing and quantity.
- **Named evidence:** The 2020 Prelims route uses tensiometer as an eco-friendly farm-water-management concept.
- **Analysis:** Measurement aligns irrigation with crop need rather than a fixed calendar.
- **Qualification / demand link:** Soil type, rooting depth, placement and instrument range affect interpretation.

EVIDENCE AND INFERENCE

- **Fact/evidence:** The 2020 Prelims route uses tensiometer as an eco-friendly farm-water-management concept.
- **Analytical inference:** Measurement aligns irrigation with crop need rather than a fixed calendar.
- **Qualification:** Soil type, rooting depth, placement and instrument range affect interpretation.

PRELIMS TRAP

Do not call a tensiometer a rainfall gauge or water pump.

MAINS USE

Use it as a precision-decision tool, not a supply source.

MINI RECAP

- Definition: A tensiometer measures soil-water tension and helps schedule irrigation in its suitable measurement range.
- Evidence anchor: The 2020 Prelims route uses tensiometer as an eco-friendly farm-water-management concept.
- Limitation: Soil type, rooting depth, placement and instrument range affect interpretation.

CORE SESSION 12 - Rebound effect in water use

VISUAL FIRST

LESS WATER PER HECTARE -> LOWER PRIVATE COST
-> MORE AREA / THIRSTIER CROP -> TOTAL EXTRACTION MAY RISE

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

The rebound effect occurs when efficiency lowers unit water cost and induces expansion that offsets expected total saving.

ANSWER-GRABBING LINE: Plot-level micro-irrigation success does not automatically prove aquifer or basin conservation.

MUST-WRITE KEYWORDS: rebound, effect, water, plot-level, micro-irrigation, success, does, automatically, prove, aquifer, basin.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Plot-level micro-irrigation success does not automatically prove aquifer or basin conservation.
- **Named evidence:** The advanced canonical owner explicitly distinguishes field saving from basin extraction.
- **Analysis:** Farmers may irrigate more area or intensify cropping after adopting drip.
- **Qualification / demand link:** Caps, metering, crop incentives and collective aquifer rules determine total use.

EVIDENCE AND INFERENCE

- **Fact/evidence:** The advanced canonical owner explicitly distinguishes field saving from basin extraction.
- **Analytical inference:** Farmers may irrigate more area or intensify cropping after adopting drip.
- **Qualification:** Caps, metering, crop incentives and collective aquifer rules determine total use.

PRELIMS TRAP

Do not equate litres saved per hectare with groundwater recovered.

MAINS USE

Pair efficiency subsidy with a resource-governance condition.

MINI RECAP

- Definition: The rebound effect occurs when efficiency lowers unit water cost and induces expansion that offsets expected total saving.
- Evidence anchor: The advanced canonical owner explicitly distinguishes field saving from basin extraction.
- Limitation: Caps, metering, crop incentives and collective aquifer rules determine total use.

CORE SESSION 13 - Groundwater-energy-crop nexus

VISUAL FIRST

CHEAP POWER -> LOW MARGINAL PUMPING COST

-> WATER-INTENSIVE CROP / DEEPER PUMPING -> AQUIFER STRESS

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

The groundwater-energy-crop nexus links electricity pricing, pumping incentives, crop returns and aquifer depletion.

ANSWER-GRABBING LINE: An electricity subsidy is simultaneously an income support and a resource-use signal.

MUST-WRITE KEYWORDS: groundwater-energy-crop, nexus, electricity, subsidy, simultaneously, income, support, resource-use, signal.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** An electricity subsidy is simultaneously an income support and a resource-use signal.
- **Named evidence:** Groundwater depletion PYQ routes and Atal Jal make behavioural governance central.
- **Analysis:** Metering, feeder reform and crop diversification can change incentives.
- **Qualification / demand link:** Poorly designed pricing can hurt small farmers without creating alternatives.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Groundwater depletion PYQ routes and Atal Jal make behavioural governance central.
- **Analytical inference:** Metering, feeder reform and crop diversification can change incentives.
- **Qualification:** Poorly designed pricing can hurt small farmers without creating alternatives.

PRELIMS TRAP

Do not treat free power as environmentally neutral or pricing as sufficient alone.

MAINS USE

Balance lifeline access, measurement, crop support and aquifer budgets.

MINI RECAP

- Definition: The groundwater-energy-crop nexus links electricity pricing, pumping incentives, crop returns and aquifer depletion.
- Evidence anchor: Groundwater depletion PYQ routes and Atal Jal make behavioural governance central.
- Limitation: Poorly designed pricing can hurt small farmers without creating alternatives.

CORE SESSION 14 - Canal externalities and drainage

VISUAL FIRST

EXCESS / UNEVEN WATER -> WATERLOGGING -> SALINITY
TAIL SCARCITY -> INEQUITY | SEEPAGE -> LOSS OR RECHARGE

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Canal externalities include waterlogging, salinity, seepage, drainage burdens and unequal delivery.

ANSWER-GRABBING LINE: Irrigation expansion must be judged by land and water quality, not only command-area hectares.

MUST-WRITE KEYWORDS: canal, externalities, drainage, irrigation, expansion, must, judged, land, water, quality, only.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Irrigation expansion must be judged by land and water quality, not only command-area hectares.
- **Named evidence:** CADWM and drainage works address field distribution and excess-water management.
- **Analysis:** Pricing and scheduling can reduce over-application at the head and improve tail supply.
- **Qualification / demand link:** Seepage effects differ across geology and may support recharge.

EVIDENCE AND INFERENCE

- **Fact/evidence:** CADWM and drainage works address field distribution and excess-water management.
- **Analytical inference:** Pricing and scheduling can reduce over-application at the head and improve tail supply.
- **Qualification:** Seepage effects differ across geology and may support recharge.

PRELIMS TRAP

Do not label every seepage flow a social loss.

MAINS USE

Use positive and negative externalities in project evaluation.

MINI RECAP

- Definition: Canal externalities include waterlogging, salinity, seepage, drainage burdens and unequal delivery.
- Evidence anchor: CADWM and drainage works address field distribution and excess-water management.
- Limitation: Seepage effects differ across geology and may support recharge.

CORE SESSION 15 - Water pricing

VISUAL FIRST

VOLUMETRIC / AREA / CROP CHARGE -> O&M FUNDING
+ DEMAND SIGNAL -> EQUITY / MEASUREMENT TRADE-OFF

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Irrigation pricing recovers some service cost and can signal scarcity, but its effect depends on measurement and alternatives.

ANSWER-GRABBING LINE: A price is incentive-compatible only when supply is reliable and users can adjust demand.

MUST-WRITE KEYWORDS: water, pricing, price, incentive-compatible, only, when, supply, reliable, users, adjust, demand.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** A price is incentive-compatible only when supply is reliable and users can adjust demand.
- **Named evidence:** State irrigation charges and electricity tariffs shape the private cost of water.
- **Analysis:** Volumetric measurement can reward conservation better than flat acreage charges.
- **Qualification / demand link:** High charges without reliable service or safeguards can exclude small and tail-end farmers.

EVIDENCE AND INFERENCE

- **Fact/evidence:** State irrigation charges and electricity tariffs shape the private cost of water.
- **Analytical inference:** Volumetric measurement can reward conservation better than flat acreage charges.
- **Qualification:** High charges without reliable service or safeguards can exclude small and tail-end farmers.

PRELIMS TRAP

Do not equate full-cost pricing with good water governance.

MAINS USE

Combine transparent tariff, service standard, lifeline protection and accountability.

MINI RECAP

- Definition: Irrigation pricing recovers some service cost and can signal scarcity, but its effect depends on measurement and alternatives.
- Evidence anchor: State irrigation charges and electricity tariffs shape the private cost of water.
- Limitation: High charges without reliable service or safeguards can exclude small and tail-end farmers.

CORE SESSION 16 - Participatory irrigation management

VISUAL FIRST

WATER USERS ASSOCIATION -> ALLOCATION / SCHEDULE / FEE
-> MAINTENANCE -> LOCAL ACCOUNTABILITY

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Participatory irrigation management gives organised users roles in distribution, maintenance and sometimes fee collection.

ANSWER-GRABBING LINE: Local participation can improve information and ownership but does not erase power asymmetry.

MUST-WRITE KEYWORDS: participatory, irrigation, management, local, participation, improve, information, ownership, does, erase, power.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Local participation can improve information and ownership but does not erase power asymmetry.
- **Named evidence:** Water Users Associations are a named institutional route in command-area reform.
- **Analysis:** Collective scheduling can reduce conflict and improve tail-end transparency.
- **Qualification / demand link:** Elite capture, unclear authority and insufficient finance can weaken associations.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Water Users Associations are a named institutional route in command-area reform.
- **Analytical inference:** Collective scheduling can reduce conflict and improve tail-end transparency.
- **Qualification:** Elite capture, unclear authority and insufficient finance can weaken associations.

PRELIMS TRAP

Do not assume creation of a WUA proves participation.

MAINS USE

Assess membership, authority, finance, inclusion and performance.

MINI RECAP

- Definition: Participatory irrigation management gives organised users roles in distribution, maintenance and sometimes fee collection.
- Evidence anchor: Water Users Associations are a named institutional route in command-area reform.
- Limitation: Elite capture, unclear authority and insufficient finance can weaken associations.

CORE SESSION 17 - PMKSY current architecture

VISUAL FIRST

PMKSY: AIBP + HKKP + WDC

PDMC -> RKVY FROM 2022-23

2021-26 APPROVAL WINDOW -> 2026 OFFICIAL CONTINUATION CONTEXT

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

PMKSY is a convergence framework for irrigation access, project completion, command management and watershed development.

ANSWER-GRABBING LINE: Component ownership and programme period must be dated because the architecture has changed.

MUST-WRITE KEYWORDS: pmksy, current, architecture, component, ownership, programme, period, must, dated, because, changed.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Component ownership and programme period must be dated because the architecture has changed.
- **Named evidence:** The 2021-26 approval covered AIBP, HKKP and WDC; official June 2026 material continues the umbrella context, while PDMC has operated under RKVY since 2022-23.
- **Analysis:** Distinct departments address major systems, minor works, watershed treatment and on-farm micro-irrigation.
- **Qualification / demand link:** An umbrella label does not make funding rules or implementing ministries identical.

EVIDENCE AND INFERENCE

- **Fact/evidence:** The 2021-26 approval covered AIBP, HKKP and WDC; official June 2026 material continues the umbrella context, while PDMC has operated under RKVY since 2022-23.
- **Analytical inference:** Distinct departments address major systems, minor works, watershed treatment and on-farm micro-irrigation.
- **Qualification:** An umbrella label does not make funding rules or implementing ministries identical.

PRELIMS TRAP

Do not continue to list PDMC as an unchanged PMKSY component after its shift.

MAINS USE

Date both the approval period and current component location.

MINI RECAP

- Definition: PMKSY is a convergence framework for irrigation access, project completion, command management and watershed development.
- Evidence anchor: The 2021-26 approval covered AIBP, HKKP and WDC; official June 2026 material continues the umbrella context, while PDMC has operated under RKVY since 2022-23.
- Limitation: An umbrella label does not make funding rules or implementing ministries identical.

CORE SESSION 18 - AIBP and Har Khet Ko Pani

VISUAL FIRST

AIBP -> COMPLETE MAJOR / MEDIUM PROJECTS
HKKP -> CADWM + SMI + RRR + BOUNDED GW WORKS

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

AIBP supports completion of major and medium irrigation projects; HKKP expands and improves field access through specified sub-components.

ANSWER-GRABBING LINE: Project funding and last-mile access solve different gaps.

MUST-WRITE KEYWORDS: aibp, khet, pani, project, funding, last-mile, access, solve, different, gaps.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Project funding and last-mile access solve different gaps.
- **Named evidence:** Official 2021-26 PMKSY design lists CADWM, surface minor irrigation and repair, renovation and restoration under HKKP, with groundwater work bounded by approvals.
- **Analysis:** Completion reduces sunk-project delay while HKKP addresses distribution and local systems.
- **Qualification / demand link:** New groundwater supply is inappropriate in over-exploited aquifers.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Official 2021-26 PMKSY design lists CADWM, surface minor irrigation and repair, renovation and restoration under HKKP, with groundwater work bounded by approvals.
- **Analytical inference:** Completion reduces sunk-project delay while HKKP addresses distribution and local systems.
- **Qualification:** New groundwater supply is inappropriate in over-exploited aquifers.

PRELIMS TRAP

Do not merge AIBP with micro-irrigation subsidy.

MAINS USE

Map each component to its scale and implementing department.

MINI RECAP

- Definition: AIBP supports completion of major and medium irrigation projects; HKKP expands and improves field access through specified sub-components.
- Evidence anchor: Official 2021-26 PMKSY design lists CADWM, surface minor irrigation and repair, renovation and restoration under HKKP, with groundwater work bounded by approvals.
- Limitation: New groundwater supply is inappropriate in over-exploited aquifers.

CORE SESSION 19 - Per Drop More Crop current location

VISUAL FIRST

MICRO-IRRIGATION SUPPORT -> PDMC
PMKSY UNTIL 2021 -> RKVY FROM 2022-23

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Per Drop More Crop is the named micro-irrigation support component now implemented under Rashtriya Krishi Vikas Yojana.

ANSWER-GRABBING LINE: The policy objective continues, but the administrative umbrella changed.

MUST-WRITE KEYWORDS: drop, more, crop, current, location, policy, objective, continues, administrative, umbrella, changed.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** The policy objective continues, but the administrative umbrella changed.
- **Named evidence:** Official Agriculture Ministry material places PDMC under RKVY from 2022-23.
- **Analysis:** State plans support drip, sprinkler and associated on-farm systems subject to guidelines.
- **Qualification / demand link:** Subsidised installation does not guarantee maintenance or basin saving.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Official Agriculture Ministry material places PDMC under RKVY from 2022-23.
- **Analytical inference:** State plans support drip, sprinkler and associated on-farm systems subject to guidelines.
- **Qualification:** Subsidised installation does not guarantee maintenance or basin saving.

PRELIMS TRAP

Do not use old PMKSY placement as the current administrative fact.

MAINS USE

State both the objective and dated scheme location.

MINI RECAP

- Definition: Per Drop More Crop is the named micro-irrigation support component now implemented under Rashtriya Krishi Vikas Yojana.
- Evidence anchor: Official Agriculture Ministry material places PDMC under RKVY from 2022-23.
- Limitation: Subsidised installation does not guarantee maintenance or basin saving.

CORE SESSION 20 - Seed multiplication chain

VISUAL FIRST

NUCLEUS -> BREEDER (GOLDEN YELLOW)
-> FOUNDATION (WHITE) -> CERTIFIED (BLUE) -> FARMER

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

The formal seed chain multiplies genetically identified material through breeder, foundation and certified classes.

ANSWER-GRABBING LINE: Each class preserves identity and quality while expanding volume for cultivation.

MUST-WRITE KEYWORDS: seed, multiplication, chain, each, class, preserves, identity, quality, while, expanding, volume.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Each class preserves identity and quality while expanding volume for cultivation.
- **Named evidence:** Seeds Rules, 1968 define foundation and certified seed; official certification practice uses golden-yellow, white and blue tags respectively.
- **Analysis:** A controlled multiplication chain supports varietal purity and traceability.
- **Qualification / demand link:** Quality can deteriorate through admixture, poor storage or unauthorised multiplication.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Seeds Rules, 1968 define foundation and certified seed; official certification practice uses golden-yellow, white and blue tags respectively.
- **Analytical inference:** A controlled multiplication chain supports varietal purity and traceability.
- **Qualification:** Quality can deteriorate through admixture, poor storage or unauthorised multiplication.

PRELIMS TRAP

Do not call certified seed the direct progeny of nucleus seed in the normal chain.

MAINS USE

Use pedigree, supervising institution and tag colour carefully.

MINI RECAP

- Definition: The formal seed chain multiplies genetically identified material through breeder, foundation and certified classes.
- Evidence anchor: Seeds Rules, 1968 define foundation and certified seed; official certification practice uses golden-yellow, white and blue tags respectively.
- Limitation: Quality can deteriorate through admixture, poor storage or unauthorised multiplication.

CORE SESSION 21 - Formal and informal seed systems

VISUAL FIRST

FORMAL: BREEDING -> CERTIFICATION / LABEL -> SALE

INFORMAL: FARM-MADE -> EXCHANGE / LOCAL MARKET

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Formal systems use recognised varieties, quality control and labelled sale; informal systems rely on farm saving, exchange and local adaptation.

ANSWER-GRABBING LINE: Seed security depends on both quality assurance and access to diverse, locally suitable material.

MUST-WRITE KEYWORDS: formal, informal, seed, systems, security, depends, both, quality, assurance, access, diverse.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Seed security depends on both quality assurance and access to diverse, locally suitable material.
- **Named evidence:** Public institutions, private companies, cooperatives and farmer networks supply different seed channels.
- **Analysis:** Formal seed can spread improved traits, while local systems preserve access and adaptation.
- **Qualification / demand link:** Informal seed may lack verified germination, while formal markets may be costly or narrow diversity.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Public institutions, private companies, cooperatives and farmer networks supply different seed channels.
- **Analytical inference:** Formal seed can spread improved traits, while local systems preserve access and adaptation.
- **Qualification:** Informal seed may lack verified germination, while formal markets may be costly or narrow diversity.

PRELIMS TRAP

Do not equate informal seed with illegal or necessarily inferior seed.

MAINS USE

Balance quality, affordability, diversity and farmer choice.

MINI RECAP

- Definition: Formal systems use recognised varieties, quality control and labelled sale; informal systems rely on farm saving, exchange and local adaptation.
- Evidence anchor: Public institutions, private companies, cooperatives and farmer networks supply different seed channels.
- Limitation: Informal seed may lack verified germination, while formal markets may be costly or narrow diversity.

CORE SESSION 22 - Seed law and current Bill status

VISUAL FIRST

SEEDS ACT 1966 + SEEDS RULES 1968
+ SEEDS CONTROL ORDER 1983
DRAFT SEEDS BILL 2025 -> CONSULTATION, NOT ENACTED BY CUTOFF

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

The Seeds Act, 1966 and Seeds Rules, 1968 remain the core quality-regulation framework for notified seed sale.

ANSWER-GRABBING LINE: A draft Bill has no legal force until enacted and commenced.

MUST-WRITE KEYWORDS: seed, current, bill, status, draft, legal, force, until, enacted, commenced.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** A draft Bill has no legal force until enacted and commenced.
- **Named evidence:** The Agriculture Ministry circulated a draft Seeds Bill, 2025 for comments in November-December 2025; no replacement Act was identified by the 9 September 2026 cutoff.
- **Analysis:** The draft sought updated registration, quality and accountability architecture.
- **Qualification / demand link:** Stakeholder concerns and legislative processing can change final provisions.

EVIDENCE AND INFERENCE

- **Fact/evidence:** The Agriculture Ministry circulated a draft Seeds Bill, 2025 for comments in November-December 2025; no replacement Act was identified by the 9 September 2026 cutoff.
- **Analytical inference:** The draft sought updated registration, quality and accountability architecture.
- **Qualification:** Stakeholder concerns and legislative processing can change final provisions.

PRELIMS TRAP

Do not present the 2019 or 2025 draft as current enacted law.

MAINS USE

State Act, Rules, Control Order and draft status separately.

MINI RECAP

- Definition: The Seeds Act, 1966 and Seeds Rules, 1968 remain the core quality-regulation framework for notified seed sale.
- Evidence anchor: The Agriculture Ministry circulated a draft Seeds Bill, 2025 for comments in November-December 2025; no replacement Act was identified by the 9 September 2026 cutoff.
- Limitation: Stakeholder concerns and legislative processing can change final provisions.

CORE SESSION 23 - Seed and planting-material diversity

VISUAL FIRST

HYBRID / VARIETY / FARM-MADE SEED
TISSUE CULTURE / BUD-CHIP = PROPAGATION MATERIAL ROUTES

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Seed choice includes genetic traits, duration, stress tolerance, quality and suitability, while some crops use vegetative planting material.

ANSWER-GRABBING LINE: Improved planting material raises attainable yield only with matching water, nutrients and management.

MUST-WRITE KEYWORDS: seed, planting-material, diversity, improved, planting, material, raises, attainable, yield, only, with.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Improved planting material raises attainable yield only with matching water, nutrients and management.
- **Named evidence:** Sugarcane bud-chip seedlings and tissue culture are propagation techniques highlighted by the 2020 Prelims route.
- **Analysis:** Rapid multiplication can spread uniform, disease-screened material where protocols work.
- **Qualification / demand link:** Narrow genetic dependence and poor-quality material increase systemic vulnerability.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Sugarcane bud-chip seedlings and tissue culture are propagation techniques highlighted by the 2020 Prelims route.
- **Analytical inference:** Rapid multiplication can spread uniform, disease-screened material where protocols work.
- **Qualification:** Narrow genetic dependence and poor-quality material increase systemic vulnerability.

PRELIMS TRAP

Do not classify tissue culture or bud chips as irrigation methods.

MAINS USE

Link seed technology to agro-climate, biosecurity and farmer cost.

MINI RECAP

- Definition: Seed choice includes genetic traits, duration, stress tolerance, quality and suitability, while some crops use vegetative planting material.
- Evidence anchor: Sugarcane bud-chip seedlings and tissue culture are propagation techniques highlighted by the 2020 Prelims route.
- Limitation: Narrow genetic dependence and poor-quality material increase systemic vulnerability.

CORE SESSION 24 - N, P and K functions

VISUAL FIRST

N -> VEGETATIVE GROWTH / PROTEIN

P -> ROOTS / ENERGY TRANSFER

K -> WATER REGULATION / STRESS AND QUALITY

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Nitrogen, phosphorus and potassium are primary plant nutrients with distinct physiological roles.

ANSWER-GRABBING LINE: Balanced fertilisation follows crop need and soil supply rather than a fixed national ratio.

MUST-WRITE KEYWORDS: functions, balanced, fertilisation, follows, crop, need, soil, supply, rather, than, fixed.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Balanced fertilisation follows crop need and soil supply rather than a fixed national ratio.
- **Named evidence:** Soil testing and agronomic recommendations connect nutrient status to dose.
- **Analysis:** Balanced inputs can raise nutrient-use efficiency and yield stability.
- **Qualification / demand link:** Deficiency, excess and nutrient interaction vary by soil, crop and timing.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Soil testing and agronomic recommendations connect nutrient status to dose.
- **Analytical inference:** Balanced inputs can raise nutrient-use efficiency and yield stability.
- **Qualification:** Deficiency, excess and nutrient interaction vary by soil, crop and timing.

PRELIMS TRAP

Do not say NPK are interchangeable or that urea supplies all three.

MAINS USE

Use nutrient function before discussing subsidy distortion.

MINI RECAP

- Definition: Nitrogen, phosphorus and potassium are primary plant nutrients with distinct physiological roles.
- Evidence anchor: Soil testing and agronomic recommendations connect nutrient status to dose.
- Limitation: Deficiency, excess and nutrient interaction vary by soil, crop and timing.

CORE SESSION 25 - Nutrient imbalance

VISUAL FIRST

CHEAP N RELATIVE TO P/K/MICRONUTRIENTS

-> EXCESS N + DEFICIENCIES -> LOWER RESPONSE / POLLUTION

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Nutrient imbalance is a mismatch between crop-soil requirement and the quantity, proportion, timing or placement of nutrients applied.

ANSWER-GRABBING LINE: Subsidy incidence changes relative prices and therefore fertiliser choice.

MUST-WRITE KEYWORDS: nutrient, imbalance, subsidy, incidence, changes, relative, prices, therefore, fertiliser, choice.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Subsidy incidence changes relative prices and therefore fertiliser choice.
- **Named evidence:** The Economic Survey and fertiliser-policy analysis repeatedly flag imbalanced use and declining response.
- **Analysis:** Site-specific nutrient management can reduce waste and correct hidden deficiencies.
- **Qualification / demand link:** A national N:P:K ratio cannot prescribe every field.

EVIDENCE AND INFERENCE

- **Fact/evidence:** The Economic Survey and fertiliser-policy analysis repeatedly flag imbalanced use and declining response.
- **Analytical inference:** Site-specific nutrient management can reduce waste and correct hidden deficiencies.
- **Qualification:** A national N:P:K ratio cannot prescribe every field.

PRELIMS TRAP

Do not treat more fertiliser as automatically more fertility.

MAINS USE

Connect relative prices, soil tests and application practice.

MINI RECAP

- Definition: Nutrient imbalance is a mismatch between crop-soil requirement and the quantity, proportion, timing or placement of nutrients applied.
- Evidence anchor: The Economic Survey and fertiliser-policy analysis repeatedly flag imbalanced use and declining response.
- Limitation: A national N:P:K ratio cannot prescribe every field.

CORE SESSION 26 - Urea control and NBS

VISUAL FIRST

UREA: GOVERNMENT-CONTROLLED MRP + SEPARATE SUBSIDY
P&K: DECONTROLLED MRP + NBS PER KG N/P/K/S

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Urea pricing remains controlled, while phosphatic and potassic fertilisers receive nutrient-based subsidy under a separate policy.

ANSWER-GRABBING LINE: Different price regimes can distort the relative cost of nitrogen and other nutrients.

MUST-WRITE KEYWORDS: urea, control, different, price, regimes, distort, relative, cost, nitrogen, other, nutrients.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Different price regimes can distort the relative cost of nitrogen and other nutrients.
- **Named evidence:** NBS has applied to eligible P&K fertilisers since 2010; seasonal nutrient rates change and are not frozen here.
- **Analysis:** Nutrient-linked support gives flexibility across product grades.
- **Qualification / demand link:** Decontrolled P&K prices remain exposed to international input costs and affordability concerns.

EVIDENCE AND INFERENCE

- **Fact/evidence:** NBS has applied to eligible P&K fertilisers since 2010; seasonal nutrient rates change and are not frozen here.
- **Analytical inference:** Nutrient-linked support gives flexibility across product grades.
- **Qualification:** Decontrolled P&K prices remain exposed to international input costs and affordability concerns.

PRELIMS TRAP

Do not say urea is covered by NBS.

MAINS USE

Distinguish product MRP, nutrient subsidy and seasonal notification.

MINI RECAP

- Definition: Urea pricing remains controlled, while phosphatic and potassic fertilisers receive nutrient-based subsidy under a separate policy.
- Evidence anchor: NBS has applied to eligible P&K fertilisers since 2010; seasonal nutrient rates change and are not frozen here.
- Limitation: Decontrolled P&K prices remain exposed to international input costs and affordability concerns.

CORE SESSION 27 - Fertiliser DBT

VISUAL FIRST

RETAIL PoS SALE TO FARMER -> TRANSACTION RECORD
-> GOVERNMENT SUBSIDY RELEASE TO COMPANY

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Fertiliser DBT links subsidy payment to authenticated retail sale but pays the subsidy to manufacturers or importers, not as cash to the farmer.

ANSWER-GRABBING LINE: The beneficiary receives a subsidised retail price rather than a bank transfer.

MUST-WRITE KEYWORDS: fertiliser, beneficiary, receives, subsidised, retail, price, rather, than, bank, transfer.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** The beneficiary receives a subsidised retail price rather than a bank transfer.
- **Named evidence:** Department of Fertilizers and PIB material describe PoS-based sale validation before company reimbursement.
- **Analysis:** Transaction data can reduce ghost movement and improve supply monitoring.
- **Qualification / demand link:** Identity, connectivity, device and dealer-stock problems can still affect access.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Department of Fertilizers and PIB material describe PoS-based sale validation before company reimbursement.
- **Analytical inference:** Transaction data can reduce ghost movement and improve supply monitoring.
- **Qualification:** Identity, connectivity, device and dealer-stock problems can still affect access.

PRELIMS TRAP

Do not confuse fertiliser DBT with PM-KISAN cash transfer.

MAINS USE

Follow the money from retail sale to company subsidy.

MINI RECAP

- Definition: Fertiliser DBT links subsidy payment to authenticated retail sale but pays the subsidy to manufacturers or importers, not as cash to the farmer.
- Evidence anchor: Department of Fertilizers and PIB material describe PoS-based sale validation before company reimbursement.
- Limitation: Identity, connectivity, device and dealer-stock problems can still affect access.

CORE SESSION 28 - Soil Health Card and INM

VISUAL FIRST

SOIL SAMPLE -> TEST -> CARD / RECOMMENDATION
-> MINERAL + ORGANIC + BIOLOGICAL SOURCES -> MONITOR

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

A Soil Health Card reports measured soil parameters and crop-oriented nutrient recommendations; integrated nutrient management combines appropriate nutrient sources.

ANSWER-GRABBING LINE: Information improves decisions only when tests are reliable and recommended inputs are accessible.

MUST-WRITE KEYWORDS: soil, health, card, information, improves, decisions, only, when, tests, reliable, recommended.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Information improves decisions only when tests are reliable and recommended inputs are accessible.
- **Named evidence:** The Soil Health Card programme operates under the Soil Health and Fertility component of RKVY from 2022-23.
- **Analysis:** Targeted application can improve nutrient balance and reduce avoidable cost.
- **Qualification / demand link:** Sampling density, laboratory quality and farmer interpretation limit precision.

EVIDENCE AND INFERENCE

- **Fact/evidence:** The Soil Health Card programme operates under the Soil Health and Fertility component of RKVY from 2022-23.
- **Analytical inference:** Targeted application can improve nutrient balance and reduce avoidable cost.
- **Qualification:** Sampling density, laboratory quality and farmer interpretation limit precision.

PRELIMS TRAP

Do not treat one card as a permanent soil diagnosis.

MAINS USE

Date the sample and combine tests with local agronomy.

MINI RECAP

- Definition: A Soil Health Card reports measured soil parameters and crop-oriented nutrient recommendations; integrated nutrient management combines appropriate nutrient sources.
- Evidence anchor: The Soil Health Card programme operates under the Soil Health and Fertility component of RKVY from 2022-23.
- Limitation: Sampling density, laboratory quality and farmer interpretation limit precision.

CORE SESSION 29 - Biochar and soil amendments

VISUAL FIRST

BIOMASS -> CONTROLLED PYROLYSIS -> BIOCHAR
-> SOIL PHYSICAL / CHEMICAL / BIOLOGICAL EFFECTS

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Biochar is carbon-rich material produced by heating biomass under limited oxygen and used as a soil amendment in suitable contexts.

ANSWER-GRABBING LINE: Its agronomic and carbon effects depend on feedstock, process, soil, crop and dose.

MUST-WRITE KEYWORDS: biochar, soil, amendments, agronomic, carbon, effects, depend, feedstock, process, crop, dose.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Its agronomic and carbon effects depend on feedstock, process, soil, crop and dose.
- **Named evidence:** The 2020 Prelims route links biochar with soil microorganisms and water-retention questions.
- **Analysis:** Stable carbon and porosity can alter water and nutrient dynamics.
- **Qualification / demand link:** Poor-quality feedstock, contaminants or unsuitable dose can harm soils.

EVIDENCE AND INFERENCE

- **Fact/evidence:** The 2020 Prelims route links biochar with soil microorganisms and water-retention questions.
- **Analytical inference:** Stable carbon and porosity can alter water and nutrient dynamics.
- **Qualification:** Poor-quality feedstock, contaminants or unsuitable dose can harm soils.

PRELIMS TRAP

Do not call all burnt residue biochar or promise universal yield gain.

MAINS USE

Present biochar as context-specific amendment and carbon option.

MINI RECAP

- Definition: Biochar is carbon-rich material produced by heating biomass under limited oxygen and used as a soil amendment in suitable contexts.
- Evidence anchor: The 2020 Prelims route links biochar with soil microorganisms and water-retention questions.
- Limitation: Poor-quality feedstock, contaminants or unsuitable dose can harm soils.

CORE SESSION 30 - Nano-fertiliser evidence discipline

VISUAL FIRST

REGULATORY NOTIFICATION / MARKETING PRODUCT
!= UNIVERSAL AGRONOMIC REPLACEMENT
USE -> CROP / SOIL / DOSE / TRIAL EVIDENCE

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Nano-fertilisers deliver nutrients in nano-scale formulations, often through foliar application.

ANSWER-GRABBING LINE: Product notification and sale do not prove universal replacement of conventional fertiliser.

MUST-WRITE KEYWORDS: nano-fertiliser, evidence, discipline, product, notification, sale, prove, universal, replacement, conventional, fertiliser.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Product notification and sale do not prove universal replacement of conventional fertiliser.
- **Named evidence:** Official 2025-26 parliamentary and ICAR material supports crop- and protocol-specific use and rejects blanket replacement claims for nano urea.
- **Analysis:** Targeted foliar supplementation may improve efficiency in some contexts.
- **Qualification / demand link:** Multi-location evidence includes inconsistent yield and quality outcomes when conventional nitrogen is sharply reduced.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Official 2025-26 parliamentary and ICAR material supports crop- and protocol-specific use and rejects blanket replacement claims for nano urea.
- **Analytical inference:** Targeted foliar supplementation may improve efficiency in some contexts.
- **Qualification:** Multi-location evidence includes inconsistent yield and quality outcomes when conventional nitrogen is sharply reduced.

PRELIMS TRAP

Do not claim one bottle universally replaces a fixed bag quantity.

MAINS USE

Use approved recommendation and independent crop-soil evidence.

MINI RECAP

- Definition: Nano-fertilisers deliver nutrients in nano-scale formulations, often through foliar application.
- Evidence anchor: Official 2025-26 parliamentary and ICAR material supports crop- and protocol-specific use and rejects blanket replacement claims for nano urea.
- Limitation: Multi-location evidence includes inconsistent yield and quality outcomes when conventional nitrogen is sharply reduced.

CORE SESSION 31 - Pesticides and IPM

VISUAL FIRST

PREVENTION -> SURVEILLANCE -> THRESHOLD
-> BIOLOGICAL / CULTURAL / MECHANICAL
-> SELECTIVE CHEMICAL LAST RESORT

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Integrated Pest Management combines compatible methods and uses pesticides judiciously according to surveillance and economic thresholds.

ANSWER-GRABBING LINE: Pest control should minimise total crop, health and ecological damage rather than maximise chemical use.

MUST-WRITE KEYWORDS: pesticides, pest, control, should, minimise, total, crop, health, ecological, damage, rather.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Pest control should minimise total crop, health and ecological damage rather than maximise chemical use.
- **Named evidence:** ICAR-NCIPM guidance and the Insecticides Act, 1968 framework anchor pest-management analysis.
- **Analysis:** Rotation, resistant varieties, natural enemies and precise application can slow resistance.
- **Qualification / demand link:** Weak diagnosis or counterfeit products can worsen residues and pest resurgence.

EVIDENCE AND INFERENCE

- **Fact/evidence:** ICAR-NCIPM guidance and the Insecticides Act, 1968 framework anchor pest-management analysis.
- **Analytical inference:** Rotation, resistant varieties, natural enemies and precise application can slow resistance.
- **Qualification:** Weak diagnosis or counterfeit products can worsen residues and pest resurgence.

PRELIMS TRAP

Carbofuran, phorate and triazophos are crop-protection chemicals, not fertilisers.

MAINS USE

Use prevention, monitoring, threshold and targeted control sequence.

MINI RECAP

- Definition: Integrated Pest Management combines compatible methods and uses pesticides judiciously according to surveillance and economic thresholds.
- Evidence anchor: ICAR-NCIPM guidance and the Insecticides Act, 1968 framework anchor pest-management analysis.
- Limitation: Weak diagnosis or counterfeit products can worsen residues and pest resurgence.

CORE SESSION 32 - Mechanisation

VISUAL FIRST

POWER SOURCE + MACHINE + OPERATOR + TIMING
-> LABOUR / SPEED / PRECISION / COST EFFECT

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Farm mechanisation substitutes or complements labour and animal power through machinery and equipment.

ANSWER-GRABBING LINE: The appropriate machine depends on farm size, operation, timing, terrain and utilisation.

MUST-WRITE KEYWORDS: mechanisation, appropriate, machine, depends, farm, size, operation, timing, terrain, utilisation.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** The appropriate machine depends on farm size, operation, timing, terrain and utilisation.
- **Named evidence:** Sub-Mission on Agricultural Mechanization and State programmes support equipment access and demonstrations.
- **Analysis:** Timely sowing, precision and reduced drudgery can raise productivity.
- **Qualification / demand link:** High capital cost, fragmented land and idle capacity can make ownership inefficient.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Sub-Mission on Agricultural Mechanization and State programmes support equipment access and demonstrations.
- **Analytical inference:** Timely sowing, precision and reduced drudgery can raise productivity.
- **Qualification:** High capital cost, fragmented land and idle capacity can make ownership inefficient.

PRELIMS TRAP

Do not equate mechanisation with tractors alone or inevitable labour displacement.

MAINS USE

Assess operation-specific cost and employment effects.

MINI RECAP

- Definition: Farm mechanisation substitutes or complements labour and animal power through machinery and equipment.
- Evidence anchor: Sub-Mission on Agricultural Mechanization and State programmes support equipment access and demonstrations.
- Limitation: High capital cost, fragmented land and idle capacity can make ownership inefficient.

CORE SESSION 33 - Custom hiring centres

VISUAL FIRST

SHARED MACHINE POOL -> RENTAL BY HOUR / ACRE
-> SMALL FARM ACCESS -> HIGHER UTILISATION

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

A Custom Hiring Centre provides machinery services without requiring each farmer to own the asset.

ANSWER-GRABBING LINE: Service models can make mechanisation scale-neutral when scheduling and maintenance work.

MUST-WRITE KEYWORDS: custom, hiring, centres, service, models, make, mechanisation, scale-neutral, when, scheduling, maintenance.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Service models can make mechanisation scale-neutral when scheduling and maintenance work.
- **Named evidence:** Agriculture Ministry mechanisation programmes promote CHCs and farm-machinery banks.
- **Analysis:** Shared ownership or rental spreads fixed cost across users.
- **Qualification / demand link:** Peak-season congestion, distance and operator shortages may exclude small farmers.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Agriculture Ministry mechanisation programmes promote CHCs and farm-machinery banks.
- **Analytical inference:** Shared ownership or rental spreads fixed cost across users.
- **Qualification:** Peak-season congestion, distance and operator shortages may exclude small farmers.

PRELIMS TRAP

Do not count a sanctioned centre as timely service delivery.

MAINS USE

Measure utilisation, downtime, pricing and access.

MINI RECAP

- Definition: A Custom Hiring Centre provides machinery services without requiring each farmer to own the asset.
- Evidence anchor: Agriculture Ministry mechanisation programmes promote CHCs and farm-machinery banks.
- Limitation: Peak-season congestion, distance and operator shortages may exclude small farmers.

CORE SESSION 34 - Credit need and sources

VISUAL FIRST

PRE-SOWING COST -> CREDIT -> INPUT PURCHASE
-> HARVEST / SALE -> REPAYMENT
SHOCK -> ROLLOVER / DISTRESS RISK

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Farm credit bridges the timing gap between cultivation expenditure and uncertain harvest revenue.

ANSWER-GRABBING LINE: Credit solves liquidity, not production risk or profitability by itself.

MUST-WRITE KEYWORDS: credit, need, sources, solves, liquidity, production, risk, profitability, itself.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Credit solves liquidity, not production risk or profitability by itself.
- **Named evidence:** Institutional sources include banks, RRBs and cooperatives; non-institutional sources include moneylenders, traders, landlords and relatives.
- **Analysis:** Timely formal credit can reduce tied sale and high-cost borrowing.
- **Qualification / demand link:** Documentation, tenancy and transaction costs can exclude vulnerable cultivators.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Institutional sources include banks, RRBs and cooperatives; non-institutional sources include moneylenders, traders, landlords and relatives.
- **Analytical inference:** Timely formal credit can reduce tied sale and high-cost borrowing.
- **Qualification:** Documentation, tenancy and transaction costs can exclude vulnerable cultivators.

PRELIMS TRAP

Do not treat loan sanction as disbursement or productive use.

MAINS USE

Trace amount, timing, purpose, cost and repayment source.

MINI RECAP

- Definition: Farm credit bridges the timing gap between cultivation expenditure and uncertain harvest revenue.
- Evidence anchor: Institutional sources include banks, RRBs and cooperatives; non-institutional sources include moneylenders, traders, landlords and relatives.
- Limitation: Documentation, tenancy and transaction costs can exclude vulnerable cultivators.

CORE SESSION 35 - Commercial banks, RRBs and cooperatives

VISUAL FIRST

COMMERCIAL BANK / RRB / COOPERATIVE NETWORK
-> BRANCH / PACS-DCCB-StCB CHANNEL -> FARM BORROWER

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

India's agricultural-credit architecture combines commercial banks, Regional Rural Banks and short-term cooperative institutions.

ANSWER-GRABBING LINE: Institutional diversity expands reach but creates differences in governance, technology and balance-sheet strength.

MUST-WRITE KEYWORDS: commercial, banks, rrbs, cooperatives, institutional, diversity, expands, reach, creates, differences, governance.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Institutional diversity expands reach but creates differences in governance, technology and balance-sheet strength.
- **Named evidence:** PACS, District Central Cooperative Banks and State Cooperative Banks form the traditional short-term cooperative chain; RRBs focus on rural priority sectors.
- **Analysis:** Local knowledge can improve borrower access and monitoring.
- **Qualification / demand link:** Weak governance, regional imbalance and land-record dependence persist.

EVIDENCE AND INFERENCE

- **Fact/evidence:** PACS, District Central Cooperative Banks and State Cooperative Banks form the traditional short-term cooperative chain; RRBs focus on rural priority sectors.
- **Analytical inference:** Local knowledge can improve borrower access and monitoring.
- **Qualification:** Weak governance, regional imbalance and land-record dependence persist.

PRELIMS TRAP

Do not call DCCBs RBI subsidiaries or PACS commercial banks.

MAINS USE

Map legal form, funding, regulation and customer interface.

MINI RECAP

- Definition: India's agricultural-credit architecture combines commercial banks, Regional Rural Banks and short-term cooperative institutions.
- Evidence anchor: PACS, District Central Cooperative Banks and State Cooperative Banks form the traditional short-term cooperative chain; RRBs focus on rural priority sectors.
- Limitation: Weak governance, regional imbalance and land-record dependence persist.

CORE SESSION 36 - NABARD role

VISUAL FIRST

RBI / GOVERNMENT POLICY -> NABARD REFINANCE / DEVELOPMENT
-> RRB / COOPERATIVE / BANK CHANNEL -> RURAL ASSET

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

NABARD is the apex development financial institution for agriculture and rural development, providing refinance, development support and specified supervisory functions.

ANSWER-GRABBING LINE: NABARD strengthens rural finance architecture but is not the universal retail KCC lender.

MUST-WRITE KEYWORDS: nabard, role, strengthens, rural, finance, architecture, universal, retail, lender.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** NABARD strengthens rural finance architecture but is not the universal retail KCC lender.
- **Named evidence:** NABARD Act, refinance operations and rural infrastructure funds provide the institutional anchor.
- **Analysis:** Refinance can lengthen resources and support underserved sectors.
- **Qualification / demand link:** Ultimate credit quality depends on lending institutions and borrower projects.

EVIDENCE AND INFERENCE

- **Fact/evidence:** NABARD Act, refinance operations and rural infrastructure funds provide the institutional anchor.
- **Analytical inference:** Refinance can lengthen resources and support underserved sectors.
- **Qualification:** Ultimate credit quality depends on lending institutions and borrower projects.

PRELIMS TRAP

Do not confuse NABARD, RBI regulation and bank ownership.

MAINS USE

Separate refinance, supervision, development and direct lending.

MINI RECAP

- Definition: NABARD is the apex development financial institution for agriculture and rural development, providing refinance, development support and specified supervisory functions.
- Evidence anchor: NABARD Act, refinance operations and rural infrastructure funds provide the institutional anchor.
- Limitation: Ultimate credit quality depends on lending institutions and borrower projects.

CORE SESSION 37 - Priority-sector agriculture target

VISUAL FIRST

BANK ANBC / CEOBSE BASE -> AGRICULTURE 18%
-> NON-CORPORATE FARMERS 14% -> SMF 10%

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Priority-sector targets require specified banks to direct a share of adjusted credit exposure to eligible sectors.

ANSWER-GRABBING LINE: A target governs portfolio classification, not an individual farmer's entitlement to a loan.

MUST-WRITE KEYWORDS: priority-sector, agriculture, target, governs, portfolio, classification, individual, farmer's, entitlement, loan.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** A target governs portfolio classification, not an individual farmer's entitlement to a loan.
- **Named evidence:** RBI's 2025 PSL Directions, updated 7 August 2026, prescribe 18% for agriculture, within it 14% for non-corporate farmers and 10% for small and marginal farmers for applicable banks.
- **Analysis:** Sub-targets can focus credit on underserved producers.
- **Qualification / demand link:** Classification incentives may encourage target-chasing without resolving appraisal or access barriers.

EVIDENCE AND INFERENCE

- **Fact/evidence:** RBI's 2025 PSL Directions, updated 7 August 2026, prescribe 18% for agriculture, within it 14% for non-corporate farmers and 10% for small and marginal farmers for applicable banks.
- **Analytical inference:** Sub-targets can focus credit on underserved producers.
- **Qualification:** Classification incentives may encourage target-chasing without resolving appraisal or access barriers.

PRELIMS TRAP

Do not apply one bank-category target blindly to every institution.

MAINS USE

Date the Direction and state ANBC/CEOBSE denominator.

MINI RECAP

- Definition: Priority-sector targets require specified banks to direct a share of adjusted credit exposure to eligible sectors.
- Evidence anchor: RBI's 2025 PSL Directions, updated 7 August 2026, prescribe 18% for agriculture, within it 14% for non-corporate farmers and 10% for small and marginal farmers for applicable banks.
- Limitation: Classification incentives may encourage target-chasing without resolving appraisal or access barriers.

CORE SESSION 38 - Kisan Credit Card

VISUAL FIRST

CROP / ALLIED WORKING-CAPITAL ASSESSMENT

-> REVOLVING KCC LIMIT -> DRAWAL -> HARVEST REPAYMENT

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Kisan Credit Card is a flexible institutional-credit mechanism for crop production and eligible allied working-capital needs.

ANSWER-GRABBING LINE: KCC is a delivery product; concessional interest depends on the separate MISS terms.

MUST-WRITE KEYWORDS: kisan, credit, card, delivery, product, concessional, interest, depends, separate, miss, terms.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** KCC is a delivery product; concessional interest depends on the separate MISS terms.
- **Named evidence:** Introduced in 1998, KCC is issued through commercial banks, RRBs and cooperatives and can cover eligible farmers, including group arrangements subject to bank rules.
- **Analysis:** Revolving access reduces repeated seasonal applications.
- **Qualification / demand link:** Land records, tenancy proof, bank access and repayment stress can exclude cultivators.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Introduced in 1998, KCC is issued through commercial banks, RRBs and cooperatives and can cover eligible farmers, including group arrangements subject to bank rules.
- **Analytical inference:** Revolving access reduces repeated seasonal applications.
- **Qualification:** Land records, tenancy proof, bank access and repayment stress can exclude cultivators.

PRELIMS TRAP

Do not describe KCC as a grant or insurance policy.

MAINS USE

Separate sanctioned limit, drawal, interest benefit and repayment.

MINI RECAP

- Definition: Kisan Credit Card is a flexible institutional-credit mechanism for crop production and eligible allied working-capital needs.
- Evidence anchor: Introduced in 1998, KCC is issued through commercial banks, RRBs and cooperatives and can cover eligible farmers, including group arrangements subject to bank rules.
- Limitation: Land records, tenancy proof, bank access and repayment stress can exclude cultivators.

CORE SESSION 39 - MISS and current KCC limits

VISUAL FIRST

BUDGET 2025-26: OVERALL KCC LIMIT ANNOUNCED Rs 5 LAKH
MISS 2025-26: 7% UP TO Rs 3 LAKH -> 3% PRI -> 4% EFFECTIVE

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Modified Interest Subvention Scheme lowers eligible short-term KCC borrowing cost through bank subvention and prompt-repayment incentive.

ANSWER-GRABBING LINE: The overall KCC ceiling announcement and the dated concessional MISS slab are not the same threshold.

MUST-WRITE KEYWORDS: miss, current, limits, overall, ceiling, announcement, dated, concessional, slab, same, threshold.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** The overall KCC ceiling announcement and the dated concessional MISS slab are not the same threshold.
- **Named evidence:** The Cabinet's FY2025-26 continuation retained 7% short-term loans up to Rs 3 lakh and 3% prompt-repayment incentive; Budget 2025-26 announced an overall KCC limit increase to Rs 5 lakh.
- **Analysis:** Prompt repayment aligns subsidy with credit discipline.
- **Qualification / demand link:** A borrower above the concessional slab or in default does not automatically receive the 4% rate.

EVIDENCE AND INFERENCE

- **Fact/evidence:** The Cabinet's FY2025-26 continuation retained 7% short-term loans up to Rs 3 lakh and 3% prompt-repayment incentive; Budget 2025-26 announced an overall KCC limit increase to Rs 5 lakh.
- **Analytical inference:** Prompt repayment aligns subsidy with credit discipline.
- **Qualification:** A borrower above the concessional slab or in default does not automatically receive the 4% rate.

PRELIMS TRAP

Do not apply the Rs 5 lakh announcement to every rupee as a 4% MISS loan.

MAINS USE

Quote year, product ceiling, subsidised slab and repayment condition.

MINI RECAP

- Definition: Modified Interest Subvention Scheme lowers eligible short-term KCC borrowing cost through bank subvention and prompt-repayment incentive.
- Evidence anchor: The Cabinet's FY2025-26 continuation retained 7% short-term loans up to Rs 3 lakh and 3% prompt-repayment incentive; Budget 2025-26 announced an overall KCC limit increase to Rs 5 lakh.
- Limitation: A borrower above the concessional slab or in default does not automatically receive the 4% rate.

CORE SESSION 40 - Tenant, JLG and SHG inclusion

VISUAL FIRST

TENANT / SHARECROPPER WITHOUT CLEAR TITLE
-> GROUP GUARANTEE / ALTERNATIVE RECORD
-> CREDIT ACCESS OR EXCLUSION

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Joint Liability Groups and Self-Help Groups can support group-based credit where individual collateral or title is weak.

ANSWER-GRABBING LINE: Inclusive product design cannot substitute for credible cultivation evidence and bank incentives.

MUST-WRITE KEYWORDS: tenant, inclusion, inclusive, product, design, cannot, substitute, credible, cultivation, evidence, bank.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Inclusive product design cannot substitute for credible cultivation evidence and bank incentives.
- **Named evidence:** RBI PSL definitions recognise SHGs and JLGs of individual farmers within non-corporate-farmer lending when data conditions are met.
- **Analysis:** Peer support and group liability can lower information and collateral barriers.
- **Qualification / demand link:** Informal tenancy, gendered land records and group failure can still exclude actual cultivators.

EVIDENCE AND INFERENCE

- **Fact/evidence:** RBI PSL definitions recognise SHGs and JLGs of individual farmers within non-corporate-farmer lending when data conditions are met.
- **Analytical inference:** Peer support and group liability can lower information and collateral barriers.
- **Qualification:** Informal tenancy, gendered land records and group failure can still exclude actual cultivators.

PRELIMS TRAP

Do not say institutional credit is legally confined to landowners.

MAINS USE

Use tenancy documentation and group models as inclusion tests.

MINI RECAP

- Definition: Joint Liability Groups and Self-Help Groups can support group-based credit where individual collateral or title is weak.
- Evidence anchor: RBI PSL definitions recognise SHGs and JLGs of individual farmers within non-corporate-farmer lending when data conditions are met.
- Limitation: Informal tenancy, gendered land records and group failure can still exclude actual cultivators.

CORE SESSION 41 - Insurance principles

VISUAL FIRST

PREMIUM + RISK POOL + INSURED EVENT
-> LOSS ASSESSMENT -> CLAIM -> INDEMNITY

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Insurance pools specified uncertain losses across insured units in return for premium.

ANSWER-GRABBING LINE: Insurance transfers covered risk; it does not prevent loss or guarantee income.

MUST-WRITE KEYWORDS: insurance, principles, transfers, covered, risk, does, prevent, loss, guarantee, income.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Insurance transfers covered risk; it does not prevent loss or guarantee income.
- **Named evidence:** PMFBY and RWBCIS are subsidised public-policy insurance schemes with defined triggers and terms.
- **Analysis:** Pooling can protect working capital and credit continuity after a shock.
- **Qualification / demand link:** Adverse selection, moral hazard, correlated weather and data delay complicate design.

EVIDENCE AND INFERENCE

- **Fact/evidence:** PMFBY and RWBCIS are subsidised public-policy insurance schemes with defined triggers and terms.
- **Analytical inference:** Pooling can protect working capital and credit continuity after a shock.
- **Qualification:** Adverse selection, moral hazard, correlated weather and data delay complicate design.

PRELIMS TRAP

Do not call disaster relief, credit waiver and insurance the same instrument.

MAINS USE

Define peril, unit, sum insured, trigger and indemnity.

MINI RECAP

- Definition: Insurance pools specified uncertain losses across insured units in return for premium.
- Evidence anchor: PMFBY and RWBCIS are subsidised public-policy insurance schemes with defined triggers and terms.
- Limitation: Adverse selection, moral hazard, correlated weather and data delay complicate design.

CORE SESSION 42 - Indemnity, yield and weather index

VISUAL FIRST

INDIVIDUAL INDEMNITY -> MEASURE FARM LOSS
AREA-YIELD INDEX -> UNIT YIELD TRIGGER
WEATHER INDEX -> STATION PARAMETER TRIGGER

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Indemnity insurance compensates assessed loss, while index insurance pays from an observable area or weather trigger.

ANSWER-GRABBING LINE: Index design reduces individual verification but creates basis risk.

MUST-WRITE KEYWORDS: indemnity, yield, weather, index, design, reduces, individual, verification, creates, basis, risk.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Index design reduces individual verification but creates basis risk.
- **Named evidence:** PMFBY mainly uses area-yield assessment for widespread loss; RWBCIS uses weather parameters.
- **Analysis:** Objective triggers can speed and standardise claims.
- **Qualification / demand link:** An individual may suffer loss when the area or weather index does not trigger.

EVIDENCE AND INFERENCE

- **Fact/evidence:** PMFBY mainly uses area-yield assessment for widespread loss; RWBCIS uses weather parameters.
- **Analytical inference:** Objective triggers can speed and standardise claims.
- **Qualification:** An individual may suffer loss when the area or weather index does not trigger.

PRELIMS TRAP

Do not treat weather insurance as insurance of actual farm yield.

MAINS USE

Compare verification cost, moral hazard and basis risk.

MINI RECAP

- Definition: Indemnity insurance compensates assessed loss, while index insurance pays from an observable area or weather trigger.
- Evidence anchor: PMFBY mainly uses area-yield assessment for widespread loss; RWBCIS uses weather parameters.
- Limitation: An individual may suffer loss when the area or weather index does not trigger.

CORE SESSION 43 - Area approach and basis risk

VISUAL FIRST

INSURANCE UNIT ACTUAL YIELD < THRESHOLD
-> AREA CLAIM
INDIVIDUAL LOSS != AREA RESULT -> BASIS RISK

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

The area approach assesses widespread yield loss for a defined insurance unit rather than every farm separately.

ANSWER-GRABBING LINE: It makes mass crop insurance feasible but can mismatch individual experience.

MUST-WRITE KEYWORDS: area, approach, basis, risk, makes, mass, crop, insurance, feasible, mismatch, individual.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** It makes mass crop insurance feasible but can mismatch individual experience.
- **Named evidence:** PMFBY relies on Crop Cutting Experiments and notified insurance units for widespread yield claims.
- **Analysis:** Pooling local yield reduces individual-loss verification cost.
- **Qualification / demand link:** Sparse, delayed or noisy CCEs weaken credibility and timing.

EVIDENCE AND INFERENCE

- **Fact/evidence:** PMFBY relies on Crop Cutting Experiments and notified insurance units for widespread yield claims.
- **Analytical inference:** Pooling local yield reduces individual-loss verification cost.
- **Qualification:** Sparse, delayed or noisy CCEs weaken credibility and timing.

PRELIMS TRAP

Do not assume every damaged farm receives a claim under an area trigger.

MAINS USE

Use basis risk to explain trust and technology reform.

MINI RECAP

- Definition: The area approach assesses widespread yield loss for a defined insurance unit rather than every farm separately.
- Evidence anchor: PMFBY relies on Crop Cutting Experiments and notified insurance units for widespread yield claims.
- Limitation: Sparse, delayed or noisy CCEs weaken credibility and timing.

CORE SESSION 44 - PMFBY current architecture

VISUAL FIRST

NOTIFIED CROP + AREA + SEASON + ENROLMENT

-> FARMER PREMIUM 2% / 1.5% / 5%

-> SUBSIDISED BALANCE -> SPECIFIED-RISK CLAIM

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

PMFBY is a subsidised crop-insurance scheme for notified crops, areas, seasons and risks.

ANSWER-GRABBING LINE: Low farmer premium shares do not mean every loss or price fall is covered.

MUST-WRITE KEYWORDS: pmfby, current, architecture, farmer, premium, shares, mean, every, loss, price, fall.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Low farmer premium shares do not mean every loss or price fall is covered.
- **Named evidence:** Official August 2026 material retains maximum farmer shares of 2% for Kharif, 1.5% for Rabi food/oilseed and 5% for annual commercial/horticultural crops; enrolment has been voluntary for farmers since Kharif 2020.
- **Analysis:** Government subsidy makes actuarial cover affordable.
- **Qualification / demand link:** State participation, premium release, data and notification choices affect delivery.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Official August 2026 material retains maximum farmer shares of 2% for Kharif, 1.5% for Rabi food/oilseed and 5% for annual commercial/horticultural crops; enrolment has been voluntary for farmers since Kharif 2020.
- **Analytical inference:** Government subsidy makes actuarial cover affordable.
- **Qualification:** State participation, premium release, data and notification choices affect delivery.

PRELIMS TRAP

Do not call the 2/1.5/5 percentages claim rates or government shares.

MAINS USE

State crop-season, sum insured, peril and voluntary status.

MINI RECAP

- Definition: PMFBY is a subsidised crop-insurance scheme for notified crops, areas, seasons and risks.
- Evidence anchor: Official August 2026 material retains maximum farmer shares of 2% for Kharif, 1.5% for Rabi food/oilseed and 5% for annual commercial/horticultural crops; enrolment has been voluntary for farmers since Kharif 2020.
- Limitation: State participation, premium release, data and notification choices affect delivery.

CORE SESSION 45 - PMFBY loss stages and technology

VISUAL FIRST

PREVENTED SOWING -> STANDING CROP -> LOCALISED CALAMITY
-> POST-HARVEST WINDOW -> CLAIM METHOD

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

PMFBY uses different assessment routes for widespread yield loss and specified local or post-harvest events.

ANSWER-GRABBING LINE: Coverage depends on notified perils and reporting or assessment rules, not any event after sowing.

MUST-WRITE KEYWORDS: pmfby, loss, stages, technology, coverage, depends, notified, perils, reporting, assessment, rules.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Coverage depends on notified perils and reporting or assessment rules, not any event after sowing.
- **Named evidence:** Official scheme material uses CCE-based yield assessment and newer YES-TECH, WINDS and CROPIC technology initiatives.
- **Analysis:** Remote sensing, weather stations and geotagged evidence can improve timeliness and audit.
- **Qualification / demand link:** Technology cannot correct late State subsidy, poor notification or inaccessible grievance by itself.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Official scheme material uses CCE-based yield assessment and newer YES-TECH, WINDS and CROPIC technology initiatives.
- **Analytical inference:** Remote sensing, weather stations and geotagged evidence can improve timeliness and audit.
- **Qualification:** Technology cannot correct late State subsidy, poor notification or inaccessible grievance by itself.

PRELIMS TRAP

Do not merge prevented sowing, local loss and area-yield claims.

MAINS USE

Match loss stage to trigger, unit and evidence.

MINI RECAP

- Definition: PMFBY uses different assessment routes for widespread yield loss and specified local or post-harvest events.
- Evidence anchor: Official scheme material uses CCE-based yield assessment and newer YES-TECH, WINDS and CROPIC technology initiatives.
- Limitation: Technology cannot correct late State subsidy, poor notification or inaccessible grievance by itself.

CORE SESSION 46 - Risk layering

VISUAL FIRST

FREQUENT SMALL LOSS -> FARM PRACTICE / SAVINGS

MEDIUM LOSS -> CREDIT / INSURANCE

CATASTROPHIC LOSS -> PUBLIC RELIEF / SOCIAL PROTECTION

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Risk layering assigns different frequencies and severities to prevention, retention, insurance and public support.

ANSWER-GRABBING LINE: Insurance is efficient for some uncertain losses, not every recurring or systemic farm problem.

MUST-WRITE KEYWORDS: risk, layering, insurance, efficient, some, uncertain, losses, every, recurring, systemic, farm.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Insurance is efficient for some uncertain losses, not every recurring or systemic farm problem.
- **Named evidence:** Diversification, irrigation, advisories, KCC, PMFBY and disaster relief occupy different layers.
- **Analysis:** Layering reduces overloading one scheme with incompatible objectives.
- **Qualification / demand link:** Highly correlated climate shocks can strain insurers and public budgets.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Diversification, irrigation, advisories, KCC, PMFBY and disaster relief occupy different layers.
- **Analytical inference:** Layering reduces overloading one scheme with incompatible objectives.
- **Qualification:** Highly correlated climate shocks can strain insurers and public budgets.

PRELIMS TRAP

Do not use loan waiver as a substitute for ex ante risk design.

MAINS USE

Match tool to frequency, severity and insurability.

MINI RECAP

- Definition: Risk layering assigns different frequencies and severities to prevention, retention, insurance and public support.
- Evidence anchor: Diversification, irrigation, advisories, KCC, PMFBY and disaster relief occupy different layers.
- Limitation: Highly correlated climate shocks can strain insurers and public budgets.

CORE SESSION 47 - Conservation agriculture

VISUAL FIRST

MINIMUM SOIL DISTURBANCE + PERMANENT SOIL COVER
+ CROP DIVERSITY / ROTATION

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Conservation agriculture combines minimum disturbance, soil cover and crop diversification or rotation.

ANSWER-GRABBING LINE: It is a systems approach, not simply zero tillage.

MUST-WRITE KEYWORDS: conservation, agriculture, systems, approach, simply, zero, tillage.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** It is a systems approach, not simply zero tillage.
- **Named evidence:** The 2018 Prelims route and ICAR agronomy support these three principles.
- **Analysis:** Residue cover and timely zero-till sowing can reduce erosion, fuel and turnaround time.
- **Qualification / demand link:** Weed management, residue equipment and local soil conditions affect results.

EVIDENCE AND INFERENCE

- **Fact/evidence:** The 2018 Prelims route and ICAR agronomy support these three principles.
- **Analytical inference:** Residue cover and timely zero-till sowing can reduce erosion, fuel and turnaround time.
- **Qualification:** Weed management, residue equipment and local soil conditions affect results.

PRELIMS TRAP

Do not claim zero tillage always increases carbon or yield.

MAINS USE

Present principles, context and transition costs.

MINI RECAP

- Definition: Conservation agriculture combines minimum disturbance, soil cover and crop diversification or rotation.
- Evidence anchor: The 2018 Prelims route and ICAR agronomy support these three principles.
- Limitation: Weed management, residue equipment and local soil conditions affect results.

CORE SESSION 48 - Integrated nutrient and pest management

VISUAL FIRST

SOIL TEST + ORGANIC / BIOLOGICAL + MINERAL NUTRIENTS
PEST SURVEILLANCE + THRESHOLD + MULTIPLE CONTROLS

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

INM balances nutrient sources; IPM integrates preventive, biological, cultural, mechanical and selective chemical controls.

ANSWER-GRABBING LINE: Integration seeks adequate productivity with lower waste and ecological damage.

MUST-WRITE KEYWORDS: integrated, nutrient, pest, management, integration, seeks, adequate, productivity, with, lower, waste.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Integration seeks adequate productivity with lower waste and ecological damage.
- **Named evidence:** Soil Health Card, ICAR extension and NCIPM guidance are named institutional anchors.
- **Analysis:** Multiple tools reduce dependence on one input and slow resistance or soil decline.
- **Qualification / demand link:** Knowledge, labour, timely diagnosis and input quality constrain adoption.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Soil Health Card, ICAR extension and NCIPM guidance are named institutional anchors.
- **Analytical inference:** Multiple tools reduce dependence on one input and slow resistance or soil decline.
- **Qualification:** Knowledge, labour, timely diagnosis and input quality constrain adoption.

PRELIMS TRAP

Do not equate integrated management with zero external input.

MAINS USE

Use site-specific diagnosis and threshold-based action.

MINI RECAP

- Definition: INM balances nutrient sources; IPM integrates preventive, biological, cultural, mechanical and selective chemical controls.
- Evidence anchor: Soil Health Card, ICAR extension and NCIPM guidance are named institutional anchors.
- Limitation: Knowledge, labour, timely diagnosis and input quality constrain adoption.

CORE SESSION 49 - Organic, natural and permaculture

VISUAL FIRST

ORGANIC = STANDARD / CERTIFICATION-BASED SYSTEM

NATURAL = LOW EXTERNAL INPUT / ON-FARM BIOLOGICAL PRACTICE

PERMACULTURE = ECOLOGICAL DESIGN FRAMEWORK

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Organic farming follows defined production and certification standards; natural farming and permaculture are related but distinct approaches.

ANSWER-GRABBING LINE: Labels must not be merged because evidence, inputs, certification and markets differ.

MUST-WRITE KEYWORDS: organic, natural, permaculture, labels, must, merged, because, evidence, inputs, certification, markets.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Labels must not be merged because evidence, inputs, certification and markets differ.
- **Named evidence:** NPOP and PGS-India anchor organic certification; natural farming policy promotes locally prepared biological inputs and reduced purchased chemicals.
- **Analysis:** Reduced synthetic-input dependence may improve ecological outcomes and cost resilience.
- **Qualification / demand link:** Transition yield, labour, nutrient balance, certification and market premium vary.

EVIDENCE AND INFERENCE

- **Fact/evidence:** NPOP and PGS-India anchor organic certification; natural farming policy promotes locally prepared biological inputs and reduced purchased chemicals.
- **Analytical inference:** Reduced synthetic-input dependence may improve ecological outcomes and cost resilience.
- **Qualification:** Transition yield, labour, nutrient balance, certification and market premium vary.

PRELIMS TRAP

Do not call every non-certified low-chemical farm organic.

MAINS USE

Define standard, input rule, verification and transition.

MINI RECAP

- Definition: Organic farming follows defined production and certification standards; natural farming and permaculture are related but distinct approaches.

- Evidence anchor: NPOP and PGS-India anchor organic certification; natural farming policy promotes locally prepared biological inputs and reduced purchased chemicals.
- Limitation: Transition yield, labour, nutrient balance, certification and market premium vary.

CORE SESSION 50 - Natural Farming Mission current status

VISUAL FIRST

CABINET APPROVAL 25 NOV 2024

-> STANDALONE NMNF THROUGH 2025-26

-> POST-MARCH 2026 STATUS REQUIRES FRESH AUTHORITY

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

The National Mission on Natural Farming was approved as a standalone Centrally Sponsored Scheme for the Fifteenth Finance Commission period.

ANSWER-GRABBING LINE: A dated approval window must not be silently extended beyond its authorised period.

MUST-WRITE KEYWORDS: natural, farming, mission, current, status, dated, approval, window, must, silently, extended.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** A dated approval window must not be silently extended beyond its authorised period.
- **Named evidence:** The 25 November 2024 approval carried Rs 2,481 crore outlay through 2025-26 and cluster, farmer and bio-input support targets.
- **Analysis:** Mission design can fund training, model farms and transition support.
- **Qualification / demand link:** Long-term continuation and outcomes require post-period official evidence.

EVIDENCE AND INFERENCE

- **Fact/evidence:** The 25 November 2024 approval carried Rs 2,481 crore outlay through 2025-26 and cluster, farmer and bio-input support targets.
- **Analytical inference:** Mission design can fund training, model farms and transition support.
- **Qualification:** Long-term continuation and outcomes require post-period official evidence.

PRELIMS TRAP

Do not present launch targets as achieved outcomes or assume automatic 2026-27 extension.

MAINS USE

Use it as a dated design and flag the current-status caveat.

MINI RECAP

- Definition: The National Mission on Natural Farming was approved as a standalone Centrally Sponsored Scheme for the Fifteenth Finance Commission period.
- Evidence anchor: The 25 November 2024 approval carried Rs 2,481 crore outlay through 2025-26 and cluster, farmer and bio-input support targets.
- Limitation: Long-term continuation and outcomes require post-period official evidence.

CORE SESSION 51 - Sikkim organic transition

VISUAL FIRST

2003 RESOLUTION -> INPUT PHASE-OUT -> CERTIFICATION
-> JANUARY 2016 FULL-ORGANIC DECLARATION

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Sikkim's organic transition was a phased State policy combining input withdrawal, extension and certification.

ANSWER-GRABBING LINE: Its ecological and branding gains are evidence of possibility, not a universal replication formula.

MUST-WRITE KEYWORDS: sikkim, organic, transition, ecological, branding, gains, evidence, possibility, universal, replication, formula.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Its ecological and branding gains are evidence of possibility, not a universal replication formula.
- **Named evidence:** The State's 2003 resolution and January 2016 declaration provide the verified chronology.
- **Analysis:** Phasing allowed institutions and farmers time to adjust.
- **Qualification / demand link:** Small area, hill ecology, market access and transition costs limit comparison with intensive plains.

EVIDENCE AND INFERENCE

- **Fact/evidence:** The State's 2003 resolution and January 2016 declaration provide the verified chronology.
- **Analytical inference:** Phasing allowed institutions and farmers time to adjust.
- **Qualification:** Small area, hill ecology, market access and transition costs limit comparison with intensive plains.

PRELIMS TRAP

Do not say an abrupt national chemical ban would reproduce Sikkim.

MAINS USE

Use chronology, benefits and transferability limits.

MINI RECAP

- Definition: Sikkim's organic transition was a phased State policy combining input withdrawal, extension and certification.
- Evidence anchor: The State's 2003 resolution and January 2016 declaration provide the verified chronology.
- Limitation: Small area, hill ecology, market access and transition costs limit comparison with intensive plains.

CORE SESSION 52 - Agroforestry, millets and integrated farming

VISUAL FIRST

TREES + CROPS / LIVESTOCK | MILLETS + DRYLAND
IFS = CROPS + ALLIED ENTERPRISES + RESOURCE RECYCLING

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Agroforestry integrates woody perennials with crops or livestock; integrated farming links enterprises and resource flows; millets can support suitable dryland systems.

ANSWER-GRABBING LINE: Diversification reduces risk only when enterprises are agronomically and commercially compatible.

MUST-WRITE KEYWORDS: agroforestry, millets, integrated, farming, diversification, reduces, risk, only, when, enterprises, agronomically.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Diversification reduces risk only when enterprises are agronomically and commercially compatible.
- **Named evidence:** Rainfed Area Development under NMSA promotes Integrated Farming Systems; the 2026 objective route tests this link.
- **Analysis:** Multiple outputs, recycling and staggered income can stabilise small farms.
- **Qualification / demand link:** Management complexity, labour and markets can create new risk.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Rainfed Area Development under NMSA promotes Integrated Farming Systems; the 2026 objective route tests this link.
- **Analytical inference:** Multiple outputs, recycling and staggered income can stabilise small farms.
- **Qualification:** Management complexity, labour and markets can create new risk.

PRELIMS TRAP

Do not equate crop diversification, rotation and integrated farming.

MAINS USE

Use complementarities, cash-flow timing and resource recycling.

MINI RECAP

- Definition: Agroforestry integrates woody perennials with crops or livestock; integrated farming links enterprises and resource flows; millets can support suitable dryland systems.
- Evidence anchor: Rainfed Area Development under NMSA promotes Integrated Farming Systems; the 2026 objective route tests this link.
- Limitation: Management complexity, labour and markets can create new risk.

CORE SESSION 53 - Climate-smart and climate-resilient agriculture

VISUAL FIRST

PRODUCTIVITY + ADAPTATION + MITIGATION WHERE POSSIBLE
-> LOCATION-SPECIFIC CLIMATE-SMART PACKAGE

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Climate-smart agriculture seeks productivity, adaptation and mitigation benefits where feasible; climate-resilient agriculture emphasises capacity to withstand and recover from shocks.

ANSWER-GRABBING LINE: The approach is an objective framework, not a universal technology list.

MUST-WRITE KEYWORDS: climate-smart, climate-resilient, agriculture, approach, objective, framework, universal, technology, list.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** The approach is an objective framework, not a universal technology list.
- **Named evidence:** ICAR, ICRISAT and the former CGIAR CCAFS programme are knowledge anchors for dryland and climate-risk research.
- **Analysis:** Stress-tolerant seed, water management, advisories and diversification can reduce vulnerability.
- **Qualification / demand link:** Trade-offs arise where mitigation options lower short-run returns or shift risk.

EVIDENCE AND INFERENCE

- **Fact/evidence:** ICAR, ICRISAT and the former CGIAR CCAFS programme are knowledge anchors for dryland and climate-risk research.
- **Analytical inference:** Stress-tolerant seed, water management, advisories and diversification can reduce vulnerability.
- **Qualification:** Trade-offs arise where mitigation options lower short-run returns or shift risk.

PRELIMS TRAP

Do not call CCAFS an Indian government scheme or every technology climate-smart.

MAINS USE

State climate hazard, farm system and distributional effect.

MINI RECAP

- Definition: Climate-smart agriculture seeks productivity, adaptation and mitigation benefits where feasible; climate-resilient agriculture emphasises capacity to withstand and recover from shocks.
- Evidence anchor: ICAR, ICRISAT and the former CGIAR CCAFS programme are knowledge anchors for dryland and climate-risk research.
- Limitation: Trade-offs arise where mitigation options lower short-run returns or shift risk.

CORE SESSION 54 - Precision and digital agriculture boundary

VISUAL FIRST

SENSOR / SATELLITE / DRONE / MODEL -> DECISION
-> VARIABLE WATER / NUTRIENT / PEST ACTION
FULL DIGITAL GOVERNANCE -> TOPIC 27

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Precision agriculture uses measured spatial or temporal variability to target farm decisions and inputs.

ANSWER-GRABBING LINE: Precision can improve efficiency but is not automatically affordable, accurate or inclusive.

MUST-WRITE KEYWORDS: precision, digital, agriculture, boundary, improve, efficiency, automatically, affordable, accurate, inclusive.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Precision can improve efficiency but is not automatically affordable, accurate or inclusive.
- **Named evidence:** Tensiometers, remote sensing, weather networks and PMFBY technology tools are bounded examples.
- **Analysis:** Targeted application can reduce excess water, nutrient or pesticide use.
- **Qualification / demand link:** Data quality, connectivity, service models and vendor lock-in shape benefit.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Tensiometers, remote sensing, weather networks and PMFBY technology tools are bounded examples.
- **Analytical inference:** Targeted application can reduce excess water, nutrient or pesticide use.
- **Qualification:** Data quality, connectivity, service models and vendor lock-in shape benefit.

PRELIMS TRAP

Do not confuse digitisation with agronomic validity.

MAINS USE

Teach the farm-use mechanism here and cross-link governance to Topic 27.

MINI RECAP

- Definition: Precision agriculture uses measured spatial or temporal variability to target farm decisions and inputs.
- Evidence anchor: Tensiometers, remote sensing, weather networks and PMFBY technology tools are bounded examples.
- Limitation: Data quality, connectivity, service models and vendor lock-in shape benefit.

CORE SESSION 55 - Input-subsidy externalities

VISUAL FIRST

LOWER PRIVATE INPUT PRICE -> HIGHER USE / CROP SHIFT
-> PRODUCTIVITY + FISCAL COST + SOIL / WATER EXTERNALITY

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

An input subsidy lowers private cost but can alter technology, crop choice and resource use beyond the transfer itself.

ANSWER-GRABBING LINE: Short-run equity and food-security objectives must be weighed against long-run externalities and opportunity cost.

MUST-WRITE KEYWORDS: input-subsidy, externalities, short-run, equity, food-security, objectives, must, weighed, against, long-run, opportunity.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** Short-run equity and food-security objectives must be weighed against long-run externalities and opportunity cost.
- **Named evidence:** Urea pricing, electricity subsidy, irrigation charges, interest subvention and insurance subsidy illustrate different channels.
- **Analysis:** Targeted support can overcome liquidity and adoption barriers.
- **Qualification / demand link:** Poor relative prices can encourage nitrogen imbalance, pumping or fiscally costly lock-in.

EVIDENCE AND INFERENCE

- **Fact/evidence:** Urea pricing, electricity subsidy, irrigation charges, interest subvention and insurance subsidy illustrate different channels.
- **Analytical inference:** Targeted support can overcome liquidity and adoption barriers.
- **Qualification:** Poor relative prices can encourage nitrogen imbalance, pumping or fiscally costly lock-in.

PRELIMS TRAP

Do not add all subsidies as if their incidence and externality were identical.

MAINS USE

Analyse beneficiary, behavioural margin, externality and reform compensation.

MINI RECAP

- Definition: An input subsidy lowers private cost but can alter technology, crop choice and resource use beyond the transfer itself.
- Evidence anchor: Urea pricing, electricity subsidy, irrigation charges, interest subvention and insurance subsidy illustrate different channels.
- Limitation: Poor relative prices can encourage nitrogen imbalance, pumping or fiscally costly lock-in.

CORE SESSION 56 - Integrated productivity-equity-sustainability reform

VISUAL FIRST

RELIABLE WATER + BALANCED INPUTS + INCLUSIVE CREDIT
+ CREDIBLE INSURANCE + EXTENSION + RESOURCE GOVERNANCE
-> RESILIENT PRODUCTIVITY

Visual reading: Identify the physical resource, price signal, institution and risk boundary before drawing the conclusion.

DEFINITION

Sustainable agricultural policy maintains productive capacity while distributing access and managing environmental and climate risk.

ANSWER-GRABBING LINE: No single input or scheme can optimise productivity, equity and sustainability.

MUST-WRITE KEYWORDS: integrated, productivity-equity-sustainability, reform, single, input, scheme, optimise, productivity, equity, sustainability.

HOW TO WRITE THE ANSWER PARAGRAPH

- **Claim:** No single input or scheme can optimise productivity, equity and sustainability.
- **Named evidence:** PMKSY/RKVY, seed law, fertiliser policy, KCC/PSL, PMFBY, ICAR extension and natural-resource institutions address different constraints.
- **Analysis:** Convergence reduces gaps between asset, adoption, finance and risk protection.
- **Qualification / demand link:** Trade-offs require local evidence, farmer participation and transition support.

EVIDENCE AND INFERENCE

- **Fact/evidence:** PMKSY/RKVY, seed law, fertiliser policy, KCC/PSL, PMFBY, ICAR extension and natural-resource institutions address different constraints.
- **Analytical inference:** Convergence reduces gaps between asset, adoption, finance and risk protection.
- **Qualification:** Trade-offs require local evidence, farmer participation and transition support.

PRELIMS TRAP

Cropping-system detail stays Topic 11 and digital-governance detail Topic 27.

MAINS USE

Conclude with risk-layered, resource-accountable and inclusion-sensitive reform.

MINI RECAP

- Definition: Sustainable agricultural policy maintains productive capacity while distributing access and managing environmental and climate risk.
- Evidence anchor: PMKSY/RKVY, seed law, fertiliser policy, KCC/PSL, PMFBY, ICAR extension and natural-resource institutions address different constraints.
- Limitation: Trade-offs require local evidence, farmer participation and transition support.

Practice is maintained in the standalone workbook.

BASIC MCQS / REMEDIATION

PYQS AND ANSWER PRACTICE

Verified routed demands, objective-key discipline and original models are maintained in the standalone workbook.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

ADVANCED 1 - Common-pool groundwater

CORE MECHANISM -> DEEPER MODEL -> EMPIRICAL LIMIT -> QUALIFIED POLICY

Open access to a shared aquifer makes each pumper ignore part of the depletion cost imposed on others; measurement and collective limits complement efficient technology.

Advanced answer line: Open access to a shared aquifer makes each pumper ignore part of the depletion cost imposed on others; measurement and collective limits complement efficient technology.

ADVANCED 2 - Jevons-style irrigation rebound

CORE MECHANISM -> DEEPER MODEL -> EMPIRICAL LIMIT -> QUALIFIED POLICY

Lower water required per hectare can expand irrigated area or induce thirstier crops, so basin withdrawal may not fall.

Advanced answer line: Lower water required per hectare can expand irrigated area or induce thirstier crops, so basin withdrawal may not fall.

ADVANCED 3 - Credit rationing

CORE MECHANISM -> DEEPER MODEL -> EMPIRICAL LIMIT -> QUALIFIED POLICY

Banks may ration credit rather than charge a sufficiently high rate when collateral, information and enforcement are weak.

Advanced answer line: Banks may ration credit rather than charge a sufficiently high rate when collateral, information and enforcement are weak.

ADVANCED 4 - Basis-risk optimisation

CORE MECHANISM -> DEEPER MODEL -> EMPIRICAL LIMIT -> QUALIFIED POLICY

A finer insurance unit can reduce basis risk but raises measurement cost and susceptibility to noise or manipulation.

Advanced answer line: A finer insurance unit can reduce basis risk but raises measurement cost and susceptibility to noise or manipulation.

ADVANCED 5 - Subsidy incidence

CORE MECHANISM -> DEEPER MODEL -> EMPIRICAL LIMIT -> QUALIFIED POLICY

The legal recipient of fertiliser, electricity, credit or insurance support need not capture the full economic benefit when prices and rents adjust.

Advanced answer line: The legal recipient of fertiliser, electricity, credit or insurance support need not capture the full economic benefit when prices and rents adjust.

ADVANCED 6 - Adaptation portfolios

CORE MECHANISM -> DEEPER MODEL -> EMPIRICAL LIMIT -> QUALIFIED POLICY

Robust climate adaptation combines water, seed, soil, information, diversification, finance and social protection instead of betting on one forecast or technology.

Advanced answer line: Robust climate adaptation combines water, seed, soil, information, diversification, finance and social protection instead of betting on one forecast or technology.

CONSOLIDATED REGISTER NOTES

1. Irrigation accounting and types

- Potential created != potential utilised != reliable field delivery != water productivity.
- Net irrigated area counts land irrigated at least once; gross area counts repeated irrigated crop instances.
- Irrigation intensity = gross irrigated area / net irrigated area x 100.
- Canals need conveyance, distribution, drainage and tail-end equity; groundwater is flexible but common-pool.
- Tanks and lift systems depend on catchment, energy and maintenance.
- Command area is served by an irrigation project; watershed is a catchment draining to one outlet.
- Drip and sprinkler improve application control but may create rebound in total extraction.

2. PMKSY and governance

- PMKSY's dated 2021-26 architecture: AIBP, HKKP and WDC; official June 2026 material retains programme context.
- HKKP covers CADWM, surface minor irrigation, RRR of water bodies and bounded groundwater works.
- PDMC supports micro-irrigation under RKVY from 2022-23.
- WDC uses ridge-to-valley catchment treatment and participatory institutions.
- Water pricing must combine measurement, service reliability, equity and O&M.
- WUAs need authority, inclusion, finance and accountability.

3. Seeds

- Nucleus -> breeder -> foundation -> certified seed.
- Tags: breeder golden yellow; foundation white; certified blue.
- Formal systems use quality control and labels; informal systems include farm saving and exchange.
- Seeds Act 1966 and Rules 1968 remain; draft Seeds Bill 2025 was not enacted by cutoff.
- Tissue culture and sugarcane bud-chip settlings are propagation methods, not irrigation.

4. Fertilisers, soil and pests

- N supports vegetative/protein functions; P roots/energy; K water regulation/stress/quality.
- Urea has controlled MRP and separate subsidy; eligible P&K fertilisers use NBS.
- Fertiliser DBT pays company after authenticated PoS sale; farmer receives subsidised price.
- Soil Health Card under RKVY Soil Health and Fertility guides dated, site-specific recommendations.
- INM combines mineral, organic and biological sources; IPM uses surveillance and thresholds.
- Nano-product approval != universal replacement evidence.

5. Machinery and credit

- Mechanisation is operation-specific; CHCs spread fixed cost and improve small-farm access.
- Institutional sources: commercial banks, RRBs and cooperatives; informal sources include moneylenders and traders.
- Cooperative short-term chain: PACS -> DCCB -> State Cooperative Bank.
- NABARD provides refinance, development support and specified supervision; RBI regulates banks.
- RBI PSL 2025 Directions, updated 7 August 2026: agriculture 18%, NCF 14%, SMF 10% for applicable banks.
- KCC is the credit product; MISS is interest support.
- Budget 2025-26: Rs 5 lakh overall KCC announcement. FY2025-26 MISS: concessional 7% up to Rs 3 lakh and 3% prompt-repayment incentive.

6. Insurance

- Insurance requires premium, insured peril/unit, sum insured, trigger, assessment and claim.
- PMFBY farmer shares: 2% Kharif; 1.5% Rabi food/oilseed; 5% annual commercial/horticultural.
- Farmer enrolment voluntary since Kharif 2020.
- Widespread yield loss uses area approach/CCEs; specified local and post-harvest risks use separate routes.
- Basis risk = individual loss differs from area-yield or weather-index result.
- YES-TECH, WINDS and CROPIC improve evidence but do not replace timely funding and grievance.

7. Sustainable agriculture

- Conservation agriculture = minimum disturbance + permanent cover + crop diversity/rotation.
- Organic, natural and permaculture are distinct.
- NMNF approved 25 November 2024 through 2025-26; later continuation needs fresh authority.
- Sikkim: 2003 resolution -> phased transition -> January 2016 full-organic declaration.
- Agroforestry integrates trees; IFS integrates enterprises and resource recycling.
- Climate-smart agriculture balances productivity, adaptation and mitigation where feasible.
- Precision tools require data quality, service access and agronomic validity; full digital governance belongs to Topic 27.

8. Reform architecture

WATER BUDGET + RELIABLE DELIVERY + BALANCED INPUTS

-> INCLUSIVE CREDIT + RISK LAYERING + EXTENSION

-> PRODUCTIVITY + EQUITY + SOIL / WATER / CLIMATE RESILIENCE

- Input subsidy analysis must identify beneficiary, behavioural response, externality and transition compensation.
- Cropping-pattern detail belongs to Topic 11; market architecture Topic 13; digital agriculture Topic 27.

Final answer line: Sustainable farm productivity requires reliable water, balanced and evidence-based inputs, inclusive finance, credible risk protection and resource governance to move together.